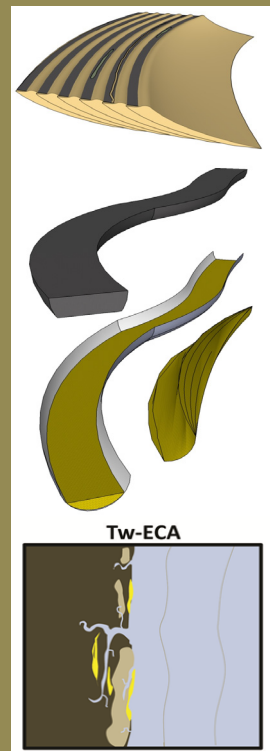
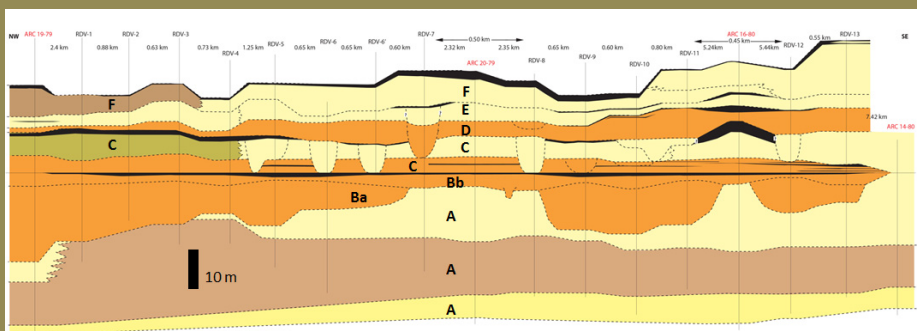
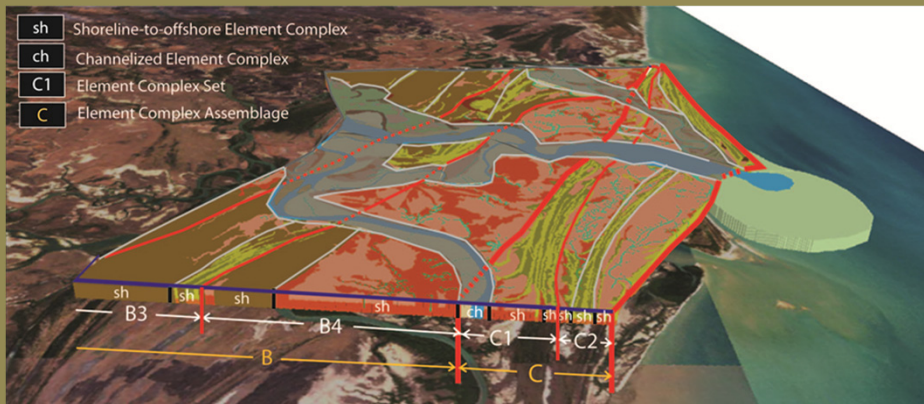


The *WAVE* Classification: A Hierarchical Approach to Process and Architectural Classification in Marginal Marine Systems



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1. SUMMARY

A new hierarchical architectural classification for elastic marginal marine depositional systems is presented and illustrated with examples. In the ancient, the architectural scheme effectively integrates the scales of sedimentology (core, outcrop) and sequence stratigraphy (wireline log correlation, reflection seismic). The classification also applies to the modern, which allows for direct comparison of architectural units between modern and ancient settings. In marginal marine systems, the parasequence typically defines reservoir flow units. This classification addresses sub-parasequence scales of stratigraphy that often control fluid flow in these reservoirs. The scheme consists of seven types of architectural units which are placed on five architectural hierarchy levels—Hierarchy Level I: Element (E) and Element Set (ES); Hierarchy Level II: Element Complex (EC) and Element Complex Set (ECS); Hierarchy Level III: Element Complex Assemblage (ECA); Hierarchy Level IV: Element Complex Assemblage Set (ECAS); and Hierarchy Level V: Transgressive-Regressive Sequence (T-R Sequence). Architectural units in Levels I to III are further classified relative to dominant depositional processes (wave, tide and fluvial) acting at the time of deposition. All architectural units are three-dimensional and can also be expressed in terms of plan-view and cross-sectional geometries. Architectural units can be linked using tree data structures by a set of familial relationships (parent-child, siblings and cousins), which provides a novel mechanism for managing uncertainty in marginal marine systems. Utilizing a hierarchical scheme permits classification of different data types at the most appropriate architectural scale. The use of the classification is illustrated in the ancient by an outcrop and subsurface example from the Campanian-aged Bearpaw-Horseshoe Canyon Formation transition, Alberta, Canada and in the modern by the Mitchell Delta, northern Australia. The case studies illustrate how the new classification can be utilized across both modern and ancient systems, in complicated, mixed-process depositional environments.

2. INTRODUCTION

Classifications attempt to introduce order to systems that otherwise display an impractical level of complexity. Classifications used in earth sciences usually involve some type of categorization, where a limited number of ‘key’ aspects of a complex, multivariate natural system are chosen as most representative. Most widely used classification schemes in sedimentary geology utilize three or four such parameters and are often illustrated with the use of 2D and pseudo 3D diagrams (e.g., Folk, 1965; Galloway, 1975; Boyd et al., 1992). The main reasons for a classification to be adopted by the scientific community are either because it is elegant and simple or, alternatively, because it offers solutions to

practical problems that other existing classifications fail to address. More than one classification scheme may be used on the same sediment body depending on the types of questions being addressed.

Classification schemes for marginal marine systems can be placed in two broad categories. The first type focuses on depositional environments (Galloway, 1975; Coleman and Wright, 1975; Hayes, 1975, 1979; Boyd et al., 1992, 2006). Such classifications are usually based on observations from modern systems, where depositional environments are comparatively easy to differentiate, with focus on the relative importance of waves, tides and fluvial influence as key parameters. Such categories are then usually linked to ancient datasets through vertical sediment log profiles and facies models (e.g., Walker and James, 1992; Posamentier and Walker, 2006). These techniques can be applied to core, wireline log motifs and outcrop successions and utilize a sedimentological, followed by a depositional environment, classification approach.

A second approach to marginal marine classification is associated with the realm of sequence stratigraphy, where stratigraphic correlation and identification of key surfaces (in both subsurface and outcrop) helps place marginal marine intervals into distinct categories (e.g., systems tracts or shoreline trajectory classes) (Posamentier et al., 1988; Van Wagoner et al., 1990; Helland-Hansen and Martinsen 1996; Helland-Hansen and Hampson, 2009). Ratios between accommodation and sediment supply (or shoreline trajectory) can be considered as the key parameters, and the resultant systems tracts (or shoreline trajectory classes) can be thought of as the resultant classification categories. Both classification approaches are often applied to the same bodies of sediments or sedimentary rocks, where a given succession can be characterized both in terms of a depositional environment and a systems tract. This is perfectly valid as long as both classification types serve a practical purpose. Often, for example, in the oil and gas industry, the sequence stratigraphic approach is useful for exploration purposes or for identification of major reservoir intervals, whereas the depositional environments approach finds direct use for reservoir development and reservoir model building.

Problems arise when the two approaches become intermixed, which is illustrated by an issue that sometimes arises when the concept of a 'parasequence' becomes associated with an environment interpretation. A parasequence is by definition a body of sediment (or sedimentary rock) bounded by flooding surfaces, internally built by genetically related beds and bedsets (Van Wagoner et al., 1990), which thus lacks any connotation to sedimentary environment. Terms such as a 'wave-dominated parasequence,' a 'deltaic parasequence,' or a 'shoreface parasequence' are therefore intermixing classifications and can lead to significant confusion, as a single parasequence can be associated with more than one depositional environment category along its down depositional dip or along depositional strike.

The frequent use of such terms in the literature does in fact point to lack of appropriate terminology that refers to scales of stratigraphy that are greater than an individual depositional environment, but yet fall within the same regressive or transgressive genetically related interval. The recent resurrection of the term ‘clinothem’ (Rich, 1951) by some workers has been an attempt of dealing with this issue in clinoform-bearing successions (Helland-Hanson, 1992; Mellere et al., 2002; Plink-Bjorklund, 2008; Bhattacharya, 2011). Another recent attempt at addressing this issue was offered by Hampson et al. (2008) for wave-dominated systems, where the authors proposed a three-fold hierarchy—(i) beach ridge, (ii) beach ridge set, and (iii) progradational wave-dominated shoreline system—the third of which corresponds to a parasequence. Beach ridges and the beach ridge sets occur on an intra-parasequence scale and, as was argued by the authors, can result in important reservoir effects. Mixed-influence systems can show significantly more variability (Bhattacharya and Giosan, 2003; Boyer et al., 2005; Gani and Bhattacharya, 2007; Plink-Bjorklund, 2008; Ainsworth et al., 2008, 2011; Vakarelov et al., 2012), where the deposits that constitute a single parasequence can represent complex juxtapositions of environments. There is, therefore, clearly a need for terminology that refers to architecture at different scales in marginal marine environments and that covers the full spectrum of process and environment variability that affects these systems.

The conceptual framework presented in this paper builds on a recent process classification scheme for marginal marine systems introduced by Ainsworth et al. (2011) (Figure 1). Ainsworth et al. (2011) focused on the process-based aspect of the classification and its predictive linkage to physiographic, accommodation/sediment supply (A/S), and other parameters. Herein, we link the process categories defined by Ainsworth et al. (2011) to resultant three-dimensional distribution of sediments (or sedimentary rocks), i.e. the stratigraphic architecture. The architectural classification presented here uses a hierarchical approach that occurs on five stratigraphic levels that describe the range of sedimentation from the scale similar to an individual lithofacies association to the scale of a T-R depositional sequence (Embry and Johannessen, 1992). The architectural classification scheme is applied equally well to both a facies analysis-based approach (depositional environments) and a regional stratigraphic approach (sequence stratigraphy), effectively linking the two. Most importantly, the classification handles the levels of stratigraphic architecture that are intermediate to those normally observed on a local level (e.g., sedimentological core description, limited outcrop exposure) and regional level (e.g., wireline log correlation and identification of key stratigraphic surfaces, regional outcrop correlation, use of seismic data). This intermediate, intra-parasequence level of stratigraphy, which is largely not dealt with in current classification schemes, is of great reservoir importance and can have a first-order control on stratigraphic reservoir compartmentalization and fluid flow within reservoirs (Figure 2) (Ainsworth, 2005;

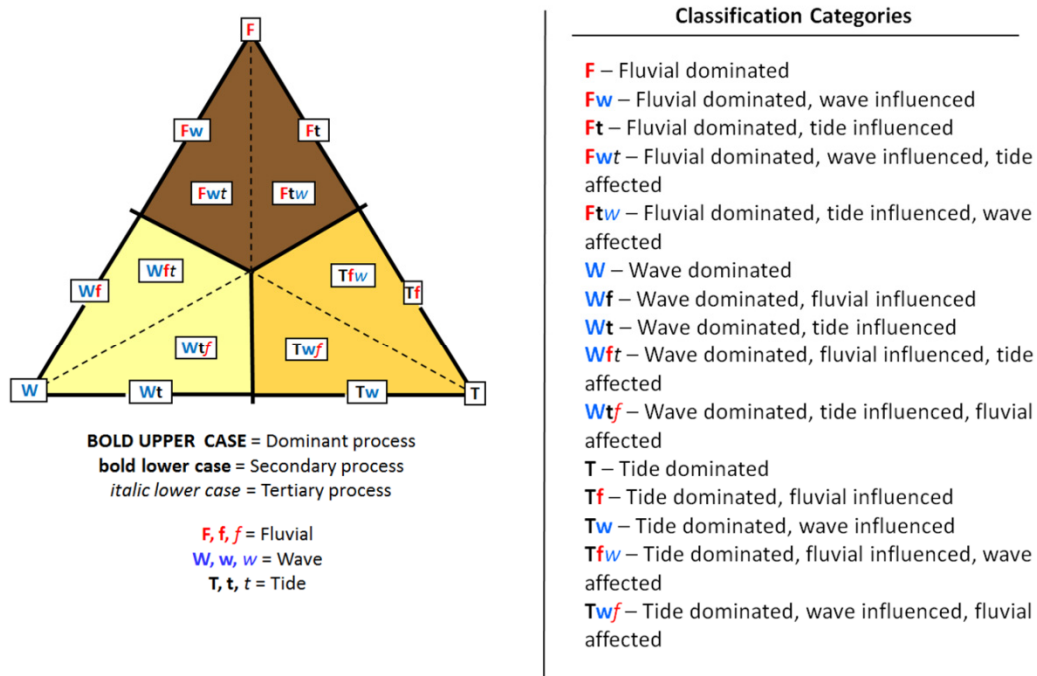


Figure 1. Process classification categories of the simple classification of Ainsworth et al. (2011). The letter codes refer to the dominant, secondary and tertiary processes (fluvial, wave and tide) that influence marginal marine systems. A Wft category, for example, refers to a wave-dominated, fluvial-influenced, tide-affected system, where the influence of waves, fluvial input and tides decreases in that order. Single letter codes refer to systems affected by a single process, two letter codes refer to systems affected by two processes, and three letter codes refer to systems affected by three processes. The classification thus effectively describes mixed influence systems. The process categories have no spatial component and can be applied to different hierarchy levels of architecture.

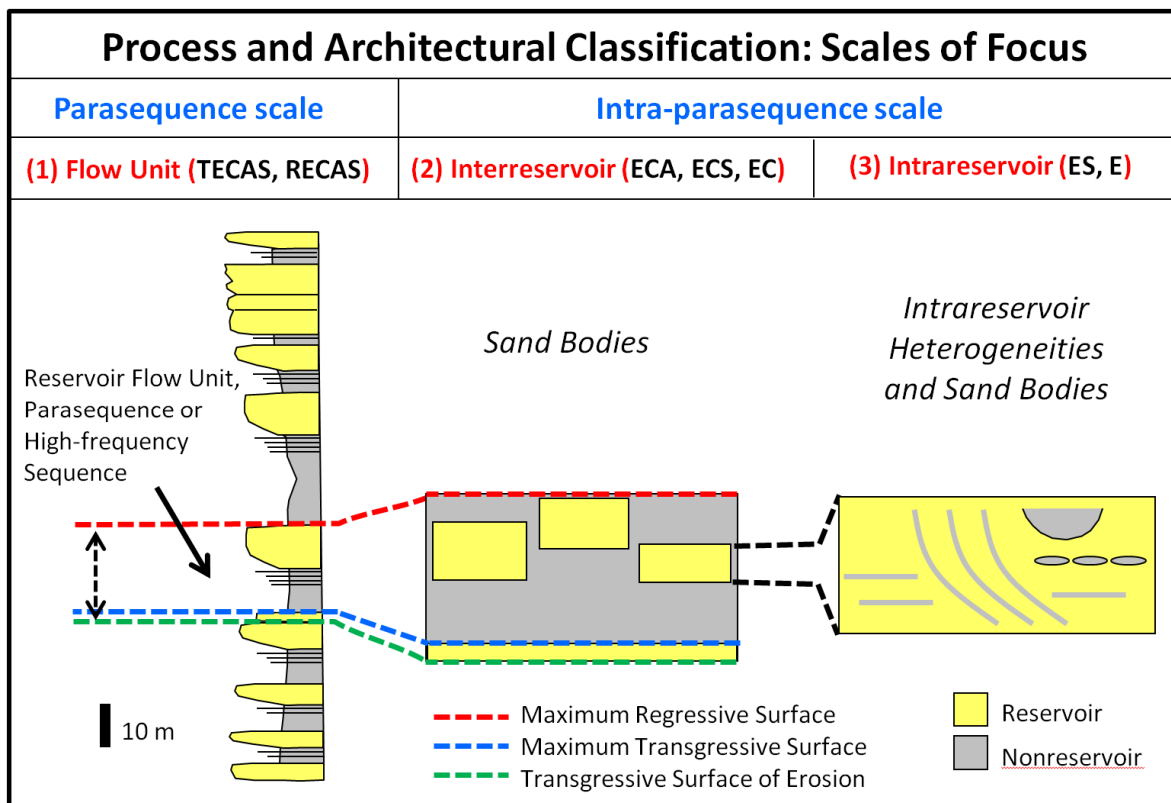


Figure 2. Hierarchical scales that influence fluid flow in marginal marine reservoirs: Flow units are usually defined by parasequence flooding surfaces, whereas inter-reservoir and intra-reservoir behavior is controlled by architecture internal to parasequences. The herein presented architectural classification describes the various types of marginal marine architecture that control the different scales of reservoir behavior. *E*, *ES*, *EC*, *ECS*, *ECA*, *RECAS*, *TECAS* and *T-R Sequence* are acronyms for new architectural categories discussed later in the text. Modified from Ainsworth (2010).

Hampson et al., 2008; Ainsworth, 2010). The proposed architectural classification scheme is all-encompassing and deals with the three coastal environment process end-members (wave, tide or fluvial-dominated systems) as well as systems influenced by two or more processes (mixed-influence systems). The classification system can be applied equally well to modern as well as ancient environments, allowing for direct comparison between the two.

The aims of this paper are:

- (1) To provide a new architectural classification for coastal depositional systems that can be applied equally well across both modern and ancient settings. This classification scheme is associated with, and is complementary to, the process-based classification presented by Ainsworth et al. (2011).
- (2) To provide examples of how the classification can be applied in practice in both modern and ancient environments, and to discuss the utility of these concepts for work in the subsurface, and especially reservoir characterization.
- (3) To discuss how the new classification can be a powerful tool for dealing with uncertainty in marginal marine interpretation, particularly in data-limited scenarios.

3. PREVIOUS CLASSIFICATIONS

Classifications of marginal marine systems are typically based on observations from modern systems, where they are categorized based on the relative importance of wave, tide and fluvial influence or other criteria (Galloway, 1975; Coleman and Wright, 1975; Hayes, 1975, 1979; Orton and Reading, 1993; Boyd et al., 1992, 2006). The most utilized classification that follows this approach is that of Galloway (1975), which focused on classification of deltas that were placed on a ternary diagram with fluvial-dominated, wave-dominated and tide-dominated end members as the apices. A number of modern deltas were then plotted based on the relative importance of the three processes. This approach was further built upon by Orton and Reading (1993), who extended the ternary diagram in the third dimension to incorporate variation in grain size. Boyd et al. (1992, 2006) combined the deltaic classification of Galloway (1975) and a wave and tide-based non-fluvial coast classification by Hayes (1975, 1979) into a classification that covers all major types of marginal-marine depositional environments (deltas, strandplains, tidal flats, estuaries and lagoons). The authors further added a dynamic component by suggesting that some of these environments occur under predominantly transgressive conditions while others occur in predominantly regressive settings. A different approach to classification was taken by Coleman and Wright (1975) who statistically evaluated parameters associated with 50 modern deltas and determined that the studied systems fall in six morphological delta types. The morphological categories of

Coleman and Wright (1975) can in fact be plotted on the Galloway (1975) ternary plot (see Bhattacharya and Walker, 1992; Bhattacharya, 2006), which nicely illustrates some of the morphological differences between systems that are fluvial-, wave- and tide-dominated, as well as intermediate, mixed influence scenarios.

The latest published classification is that of Ainsworth et al. (2011) who built on the Galloway (1975) model but directly addressed the issue of mixed-influence deltaic and non-deltaic systems by defining and naming intermediate categories (Figure 1). The classification uses a one-to-three letter code that designates the dominant, secondary, and tertiary processes that impact on a coastline. A system that is classified as W, for example, is solely wave-dominated and is not affected by fluvial or tidal influence; a system that is classified as Wft is wave-dominated, fluvial-influenced, tide-affected, where the influence of waves, fluvial input and tides decreases in that order. The categories in the simplified classification version presented in Ainsworth et al. (2011) are F, Fw, Ft, Fwt, Ftw, W, Wf, Wt, Wft, Wtf, T, Tw, Tf, Twf, and Tfw. Unlike the other classifications discussed in this section, the classification presented in Ainsworth et al. (2011) is process-based and does not directly address the morphology of sediment that forms under such conditions.

Very few attempts at describing the hierarchy of the various morphological or architectural components of marginal marine systems have previously been attempted, and these have typically been limited to individual case studies or environments. Hampson et al. (2008) suggested a three-level subdivision of wave-dominated end members ('W' category of Ainsworth et al., 2011) consisting of beach ridge, beach ridge set, and progradational wave-dominated shoreline system categories. No attempt was made by the authors to extend this approach to other marginal marine depositional environments. Gani and Bhattacharya (2007) attempted to apply a facies architecture approach to a deltaic system following a method of study usually used in fluvial and eolian investigations (Brookfield, 1977; Allen, 1983; Miall, 1985, 1996). After defining a hierarchy of bounding surfaces, the authors defined "architectural elements" that were defined as a single stratum or a package of strata that can be regarded as the smallest but meaningful unit (i.e. building block) enclosed by distinct, through-going surfaces of a depositional system." Six architectural elements were defined and mapped in a detailed outcrop investigation in the Raptor Ridge area, but no attempts were made at placing these within larger-scale architectural categories. The entire regressive system was instead referred to as the Raptor Delta. Gani and Bhattacharya (2007) show a 15km long, along-strike cross section through the Raptor delta that displays lateral changes in both primary and secondary processes as determined by sedimentary structures (Figure 3). Following the process classification of Ainsworth et al. (2011), Log A is process classified as Fw, Log B is process classified as Ftw, Log C is process classified as Wf, and D is process classified as W. The process

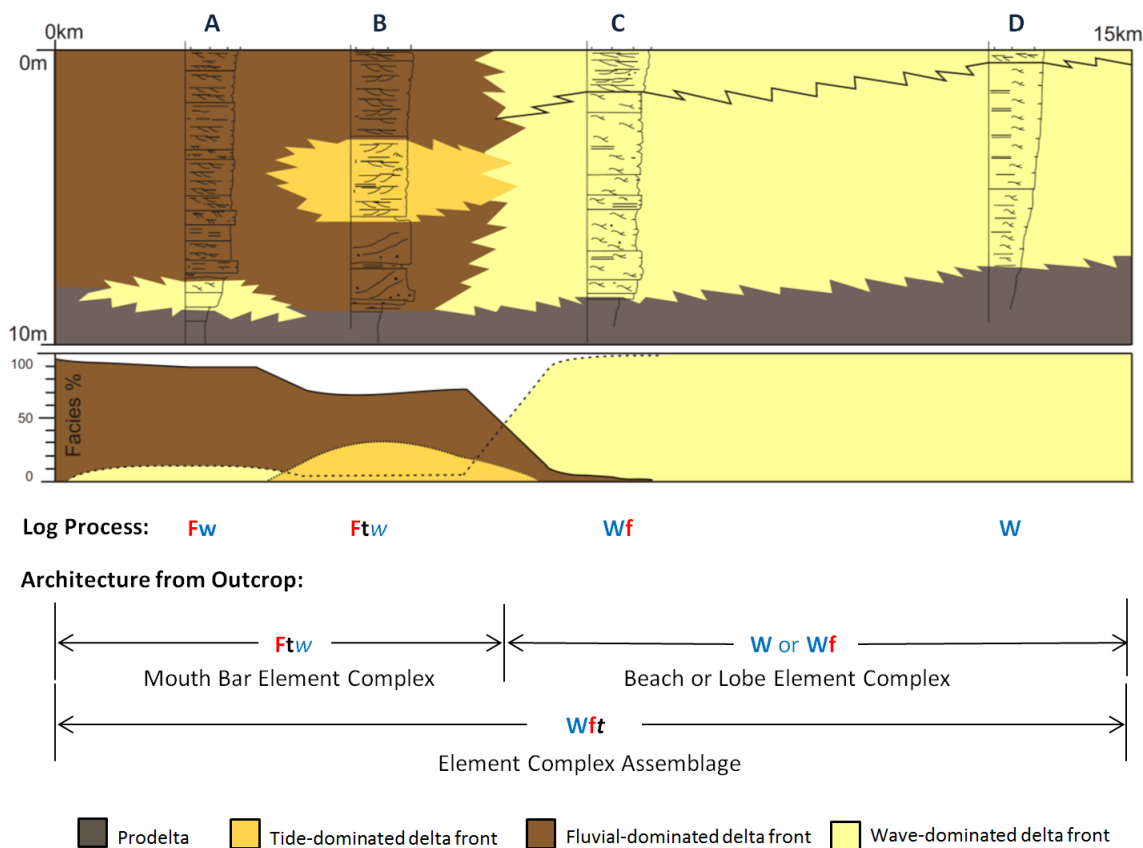


Figure 3. An along strike cross-section through the Cretaceous Raptor Delta, Wall Creek Member, Frontier Formation, Wyoming (modified from Gani and Bhattacharya, 2007) showing lateral changes in process domination in the same unit over a distance of 15km. Note the process classification can be applied to individual logs. Based on this data we can define two different Element Complexes (ECs) and a single Element Complex Assemblage (ECA) (see text for explanation of architectural units).

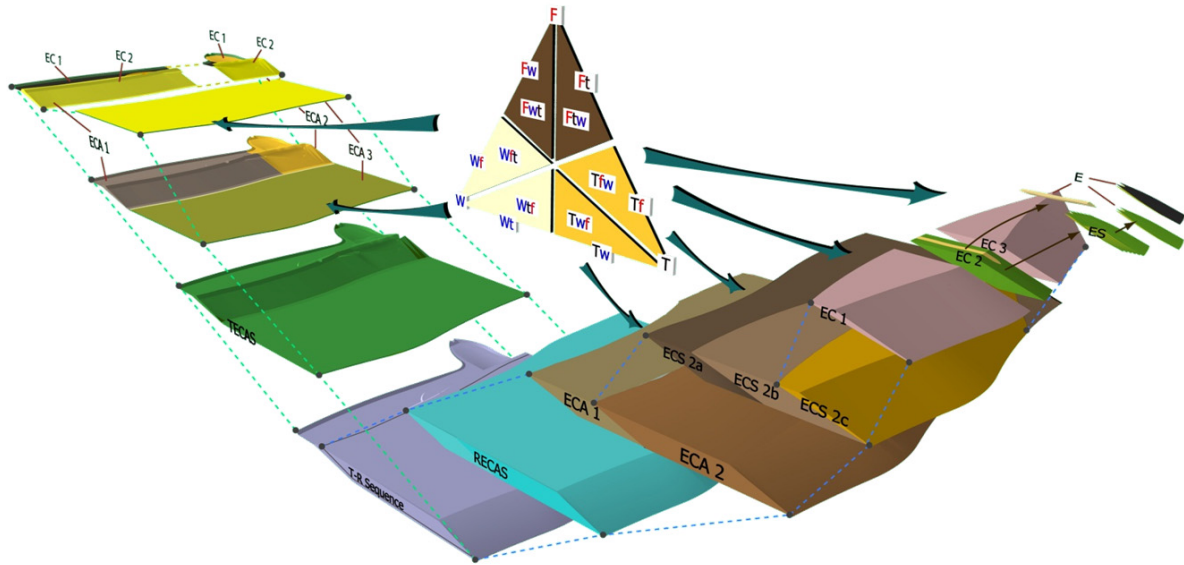
classifications are based on the proportion of facies in the sedimentary logs. The architectural description of this degree of complexity within the same regressive system (parasequence) will be the subject of the next section.

4. ARCHITECTURAL CLASSIFICATION

4.1 Hierarchy Levels and Architectural Units

A nested architectural approach that effectively incorporates different scales of marginal marine stratigraphy has the potential to resolve issues between local and regional observation and existing confusion in nomenclature. This is of great practical importance for subsurface work, where determining reasonable ranges of dimensions for geobodies, based on limited core and wireline data, is required for correlation and model building. In order to resolve this issue, we introduce a new marginal marine architectural classification that uses seven different architectural units that are placed on five stratigraphic hierarchy levels (Figures 4 and 5). The architectural units describe stratigraphic variability from the scale of an individual lithofacies to the scale of a transgressive-regressive sequence (T-R sequence; Embry and Johannessen, 1992; Embry, 1995), seamlessly integrating facies analysis with sequence stratigraphy. We define the following hierarchy levels and architectural units which increase spatially from small to large: Hierarchy Level I (Element and Element Set); Hierarchy Level II (Element Complex and Element Complex Set); Hierarchy Level III (Element Complex Assemblage); Hierarchy Level IV (Element Complex Assemblage Set); Hierarchy Level V (T-R Sequence) (Figures 5, 6 and 7). Figure 4 lists the abbreviations and suggested common names for the introduced architectural terms. All architectural units are by definition three-dimensional (Figure 4) and therefore have an expression in terms of both plan view and cross-section. This in turn allows the direct comparison between the same units in modern and ancient settings.

Hierarchy levels denote the relative positions of architectural units in the nested hierarchy, with successive levels representing increasing scales of observation. Placement of an architectural unit within a given hierarchy level is determined by its parent-child relationships relative to other architectural units in the classification. Architectural units which can only have a single other architectural unit as a parent or occupy the top level in the hierarchy (Level V) are placed in a hierarchy level of their own (e.g., Element Complex Assemblage, Element Complex Assemblage Set, T-R Sequence) (Figure 5). Architectural units that can share a common parent are placed in the same hierarchy level (e.g., Elements and Element Sets; Element Complexes and Element Complex Sets) (Figure 5).



Architectural Unit Name	Acronym	Informal Abbreviated Name
Element *	E	<i>No abbreviation</i>
Element Set *	ES	<i>No abbreviation</i>
Element Complex *	EC	Complex
Element Complex Set	ECS *	Complex Set
Element Complex Assemblage	ECA *	Assemblage
Element Complex Assemblage Set	ECAS *	Assemblage Set
Transgressive Element Complex Assemblage Set	TECAS *	Transgressive Assemblage Set
Regressive Element Complex Assemblage Set	RECAS *	Regressive Assemblage Set
Transgressive-Regressive Sequence	T-R Sequence*	<i>No abbreviation</i>

* - recommended form for usage

Figure 4. A summary diagram of the process and architectural marginal marine classification with 3D representations of the newly introduced architectural units (top) and a list of full names, acronyms, informal abbreviated names, and recommended forms for usage (bottom). The classification is hierarchical with combinations of smaller-scale architectural units building larger-scale architectural units. Green arrows refer to the architectural units that are associated with a process definition. See Figure 1 for ternary plot acronym explanations.

Figure 5. The architectural classification represented as a tree structure where individual architectural units are linked to each other by parent-child relationships. Unit EC₂, for example, has categories ES₁, E₂, and E₃ as children and ECS₁ as a parent. ECS₁ is in turn linked to ECA₂, RECAS, and T-R Sequence through a set of parent-child relationships (red arrows). The nature of the links is used to define Hierarchy Levels I - V. An architectural unit which has a parent-child relationship with a single other architectural unit is placed in its own hierarchy level (e.g., Element Complex Assemblage unit and Hierarchy Level III). Architectural units that share a parent are placed on the same hierarchy level (e.g., Element Complex and Element Complex Set units and Hierarchy Level II). See Figure 4 for a list of used architectural unit acronyms. See Tables 1-2 and Figures 6-11 for discussion of architectural units and hierarchy levels. Architectural units labeled with an 'x' subscript (E_x and EC_x) are meant to represent an unspecified architectural unit category.

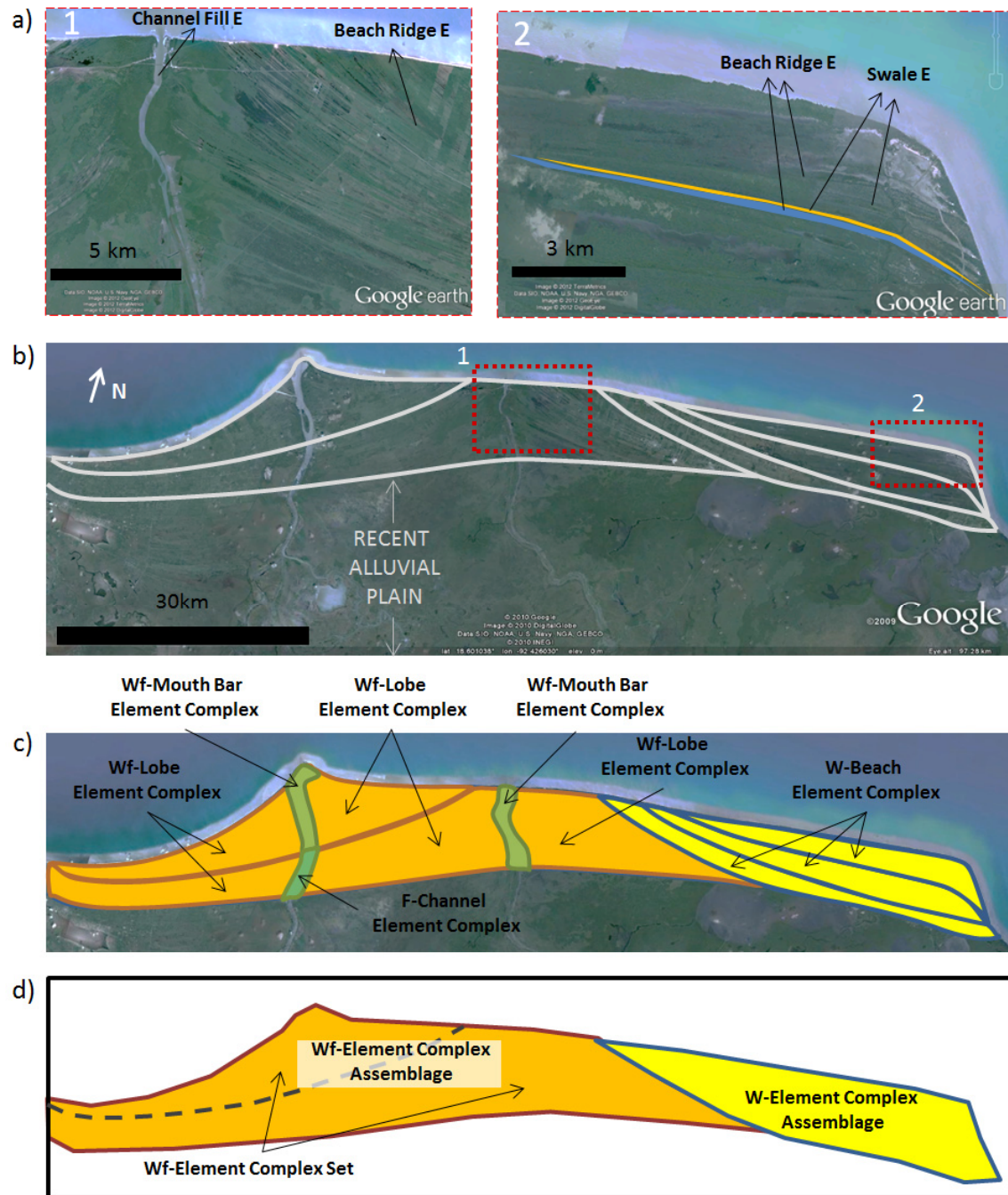


Figure 6. Hierarchy Levels I, II and III of the architectural classification as illustrated by plan view geometries in the modern Frontera system, Tabasco, Mexico. (a) Small-scale geomorphologic features, such as channels and beach ridges, form the basis for the definition of Elements. Elements are three-dimensional architectural units that are considered here as the building blocks of depositional systems. (b) The larger-scale organization of the Frontera system showing bounding surfaces (white lines) that mark periods of delta lobe and strandplain growth in the system. Red rectangles correspond to the two areas shown in (a). (c) Named Element Complexes (ECs) in the Frontera system bounded by minor bounding surfaces. Each Element Complex has a process classification (e.g., Wf) and a descriptive parameter referring to its type (e.g., Lobe, Mouth Bar, and Beach Element Complex). (d) Element Complex Sets (ECSs) and Element Complex Assemblages (ECAs) in the Frontera system (see text for discussion). See Figure 4 for a list of used architectural unit acronyms. See Tables 1-2 and Figures 7-11 for discussion of architectural units and hierarchy levels.

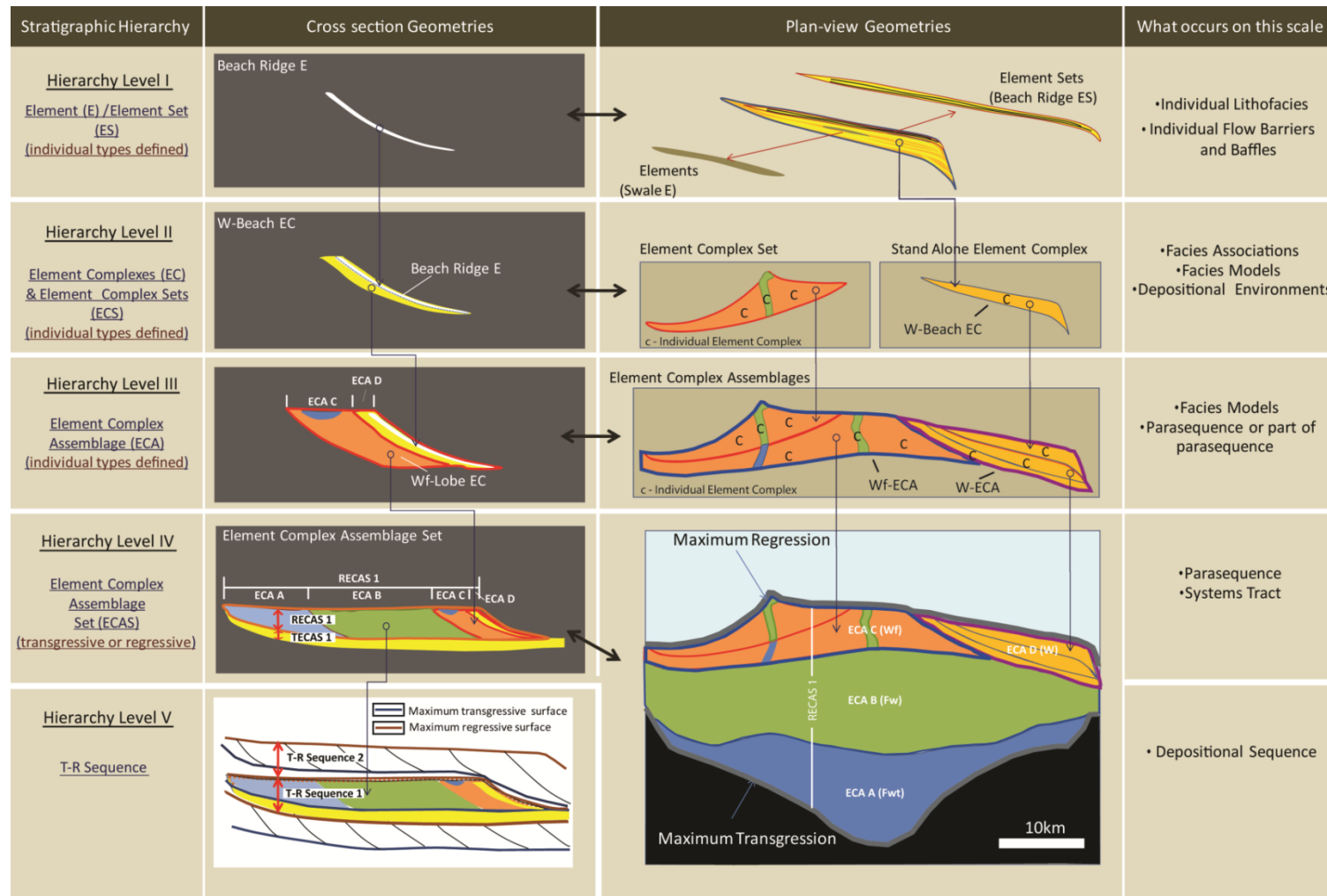


Figure 7. A diagram of the five hierarchy levels of the architectural classification showing the various architectural units in terms of cross sectional geometries (second column) and plan view geometries (third column). Architectural units cover the full spectrum of sediment bodies from an individual lithofacies to sequence stratigraphy. A lists of common sedimentological and stratigraphic terms that apply to the same scale as the hierarchy levels are shown in the fourth column. The arrows between architectural units illustrate the nested nature of the classification. See Figure 4 for explanation of acronyms used in the figure. See Table 2 for definition of architectural units.

The best way to illustrate the conceptual framework of the architectural classification, and its hierarchical nature, is with an example. Figure 6 shows how architectural unit categories belonging to Hierarchy Levels I, II and III are defined in the modern wave-dominated, fluvial-influenced Frontera coastal system, Tabasco, Mexico. Figure 6a shows two close-up areas that are marked with red rectangles in Figure 6b.

4.1.1 Hierarchy Level I – Element (E) and Element Set (ES)

Various geomorphological features such as beach ridges, swales between beach ridges, elongated swamps, as well as fluvial channels can be recognized in the Frontera system (Figure 6). We consider this level of architecture to represent the basic building blocks of marginal marine depositional systems and use it to define Elements. An Element, as we use the term, is defined as a three-dimensional sediment body that has both a surface and subsurface expression. The definition of a Beach Ridge Element for example contains its plan view subaerial expression, as well as its subsurface shoreface equivalent (Figure 8). This definition is thus markedly different from the geomorphological definition of a beach ridge (see Neuendorf et al., 2005). Using such an approach, several Element categories can be defined in the Frontera system, such as Beach Ridge Elements, Swale Elements, Channel Fill Elements, and Coastal Swamp Fill Elements (Figure 6a). Some Elements naturally arrange themselves into Element Sets that are built by Elements of the same category and orientation (see Hampson et al., 2008). Examples of Element Sets are Mouth Bar Element Sets and Beach Ridge Element Sets. We have defined and named an array of Element and Element Set categories that describe the main types of variation in marginal marine systems (Figure 8; Table 1).

4.1.2 Hierarchy Level II – Element Complex (EC) and Element Complex Set (ECS)

Elements and Element Sets arrange themselves into larger-scale architectural units that can also be named. We refer to such units as Element Complexes, which are placed in Hierarchy Level II of the architectural classification (Figure 5). Element Complexes constitute components of genetically related depositional environments formed under a given range of local wave, tide and fluvial process conditions (see Table 2). Varying the local process conditions along a shoreline or through time will result in the formation of different Element Complexes. Since Element Complexes form components of depositional systems rather than entire depositional systems, it is useful for them to be placed into distinct depositional zones (Figure 9). Element Complexes deposited at a shoreline or shallow-marine depositional setting are said to belong to a Shoreline-to-offshore Zone; Element Complexes accumulated as part of isolated channel or channel belt deposits are said to belong to a Channelized Zone; and Element Complexes associated with non-channelized onshore deposition are said to belong to an Onshore Zone (Figure 9;

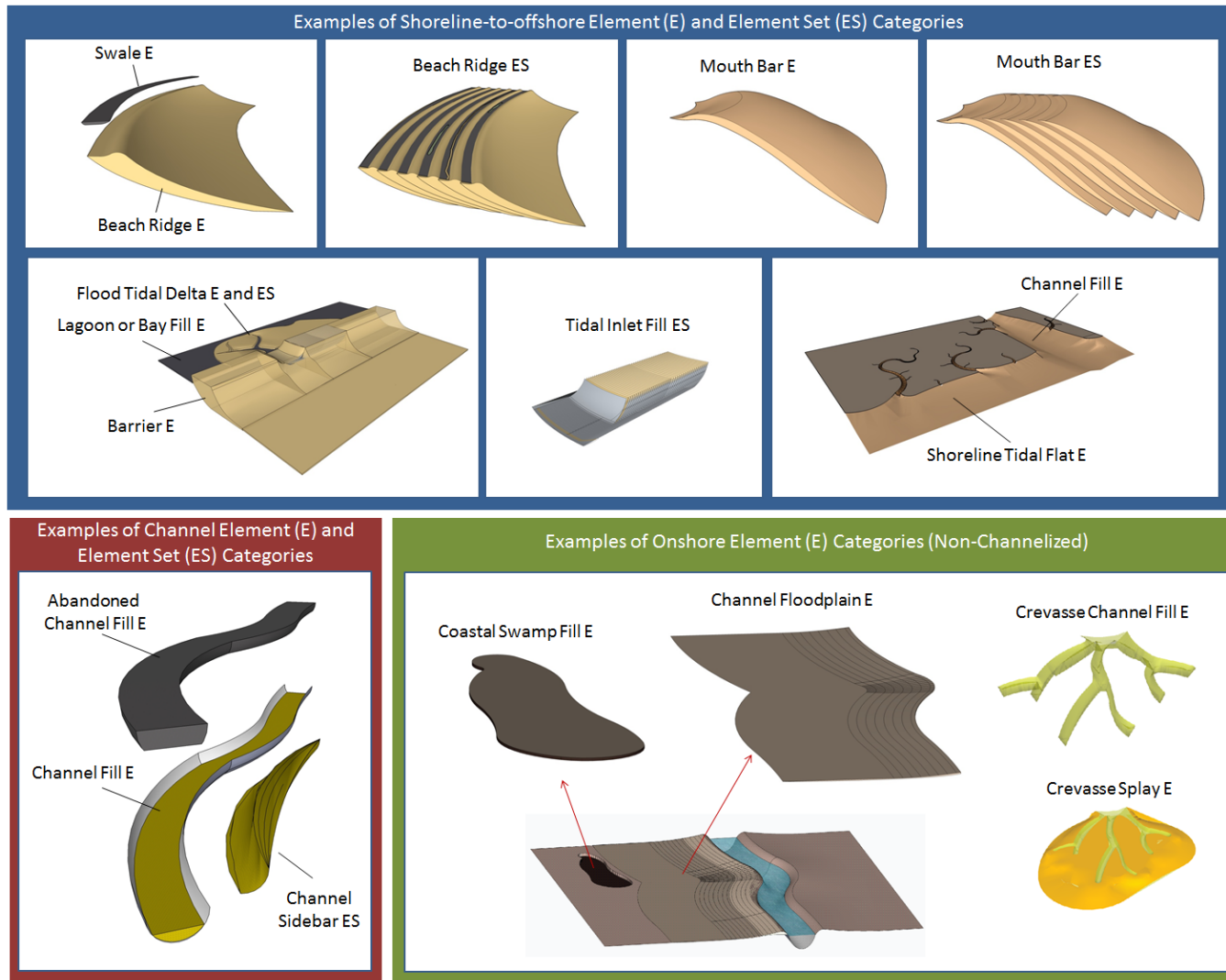


Figure 8. 3D illustrations of a sample of named Element (E) and Element Set (ES) categories in the architectural classification. Note that some Elements occur individually while others tend to arrange themselves in Sets. Elements are the building blocks of depositional systems and can be related to larger-scale features such as Element Complexes. For a complete list of Elements consult Table 1. Definition of the term Element is provided in Table 2.

Table 2. Definitions for newly introduced architectural terminology used in the text.

Term	Definition
Architectural Unit	A general term referring to the seven types of sedimentary bodies defined in the architectural classification. Architectural units are three-dimensional and can be described in terms of their three-dimensional shape or in terms of their cross-sectional and plan-view geometries. Units cover a range of scales and employ a nested architectural structure, where larger units are comprised of smaller ones. Architectural units are subdivided into a number of Architectural Unit Categories. <i>Example: Element Complex (EC)</i>
Hierarchy Level	The seven types of architectural units are placed into five parent-child based hierarchy levels (Hierarchy Levels I-V). Successive levels denote increasing scales of observation. Architectural units are placed in a level of their own when they either occupy the top node of the tree structure (T-R Sequence), or when they are the only possible child of a parent architectural unit. A level contains more than one architectural unit type if the units can share a common parent. <i>Example: Level II</i>
Process Category	One of fifteen process categories determined by ratios between fluvial, wave, and tide influence in marginal marine systems. The process categories are represented by a one to three letter code in which the first letter refers to the dominant process, the second letter refers to the secondary process, and the third letter refers to the tertiary process (see Ainsworth et al., 2011). <i>List of Process Categories: F, Fw, Ft, Fwt, Ftw, W, Wf, Wt, Wft, Wtf, T, Tw, Tf, Twf, Tfw</i>
Architectural Unit Category	Each Architectural Unit is subdivided into a set of named “Architectural Unit Categories” that describe different portions of the process and architectural spectrum typical for the unit (see Table 1). Element, Element Complex, Element Complex Set, and Element Complex Assemblage Categories are named based on a process and an architectural parameter (e.g., W-Beach Element Complex). Element Complex Assemblage Set (ECAS) and T-R Sequence Categories are not assigned a process (e.g., Regressive Element Complex Assemblage Set (RECAS)). Architectural Unit Categories are used for establishing parent-child and sibling relationships between units. <i>Example: W-Beach Element Complex</i>
Element (E)	An architectural unit that represents the smallest unique, identifiable geomorphological feature, and its subsurface equivalent, from which larger-scale architectural units are built. Elements can arrange themselves into Element Sets, or can be parts of larger groupings of Elements and Element Sets called Element Complexes. The Element architectural unit is subdivided in a number of Element categories. See Table 1 for full list of categories. <i>Example of Element category: Beach Ridge Element</i>
Element Set (ES)	An architectural unit that is formed by a group of Elements of the same type, and usually physical orientation, which share a common genetic mechanism for their formation. The Element Set architectural unit is subdivided in a number of Element Set categories. See Table 1 for full list of categories. <i>Example of Element Set category: Beach Ridge Element Set</i>
Element Complex (EC)	An architectural unit that represents a grouping of genetically-related Elements and Element Sets that were deposited under similar process conditions in the same portion of a depositional system. Element Complex boundaries can occur along strike and along dip direction and are marked by presence of distinct stratigraphic surfaces or abrupt facies boundaries. The Element Complex architectural unit is subdivided in a number of Element Complex categories. See Table 1 for full list of categories. <i>Example of Element Complex category: W-Beach Element Complex</i>
Element Complex Set (ECS)	An architectural unit that is formed by a group of genetically-related Element Complexes that represent a major phase of shoreline reorganization during regression or transgression. Element Complex Sets are always associated with a process category which is a function of regional process conditions. Element Complex Sets are only defined when evidence for shoreline reorganization internal to an Element Complex Assemblage can be identified. The Element Complex Set architectural unit is subdivided in a number of Element Complex Set categories. See Table 1 for full list of categories. <i>Example of Element Complex Set category: W-Element Complex Set</i>

Element Complex Assemblage (ECA)	An architectural unit that is formed by a group of genetically-related Element Complexes (or Element Complex Sets if such are defined) that can cumulatively be considered to have been formed under similar process conditions. Element Complex Assemblages are conceptually similar to depositional systems. Boundaries between Element Complex Assemblages are marked by surfaces that represent a change in process conditions either laterally (along the shoreline) or relative to older or younger units in up-dip or down-dip depositional direction. The process change may or may not represent a major phase of shoreline reorganization during regression or transgression. The Element Complex Assemblage architectural unit is subdivided in a number of Element Complex Assemblage categories. See Table 1 for full list of categories. <i>Example of Element Complex Assemblage category: W-Element Complex Assemblage</i>
Element Complex Assemblage Set (ECAS)	An architectural unit that represents a grouping of Element Complex Assemblages formed during a single, regional pulse of shoreline transgression or regression. The Element Complex Assemblage Set architectural unit can be subdivided into two Element Complex Assemblage Set categories, a Transgressive Element Complex Assemblage Set (TECAS) and a Regressive Element Complex Assemblage Set (RECAS). <i>Example of ECAS category: Regressive Element Complex Set (RECAS)</i>
T-R Sequence	An architectural unit that represents one transgressive and regressive cycle of deposition. It is comprised of a Transgressive Element Complex Assemblage Set (TECAS) overlain by a Regressive Element Complex Assemblage Set (RECAS). The T-R Sequence architectural unit has a single category of the same name. <i>Example of T-R Sequence Category: T-R Sequence</i>
Zone (or Zonation)	Refers to a group of Element Complex categories that are categorized based on broad depositional setting, depending on whether they were deposited in a shoreline-to-offshore environment, channelized environment, or an onshore non-channelized environment. It is often convenient to determine Zonation before a specific Element Complex category is identified. The zonation types are: (1) Shoreline-to-offshore, (2) Channelized, and (3) Onshore (see Table 1 and Figure 9).
Shoreline-to-offshore Zone	Refers to Element Complexes associated with shoreline deposition between Mean High Tide level and the toe of the shoreface (see Figures 9 and 10).
Channelized Zone	Refers to Element Complexes associated with channels and their related deposits (see Figures 9 and 10).
Onshore Zone	Refers to Element Complexes associated with brackish (between Mean High Tide and Mean Spring Tide levels) to non-marine, non-channelized deposition (see Figures 9 and 10).
Element Complex Type	Refers to a group of architecturally similar Element Complex Categories within a Zone. The following Element Complex types are defined: (i) Shoreline-to-offshore Zone: Barrier, Beach, Lagoon, Lobe, Mouth Bar, Tidal Flat, Transgressive Lag EC types; (ii) Channelized Zone: Channel and Funnel Basin EC types; (iii) Onshore Zone: Onshore EC type.

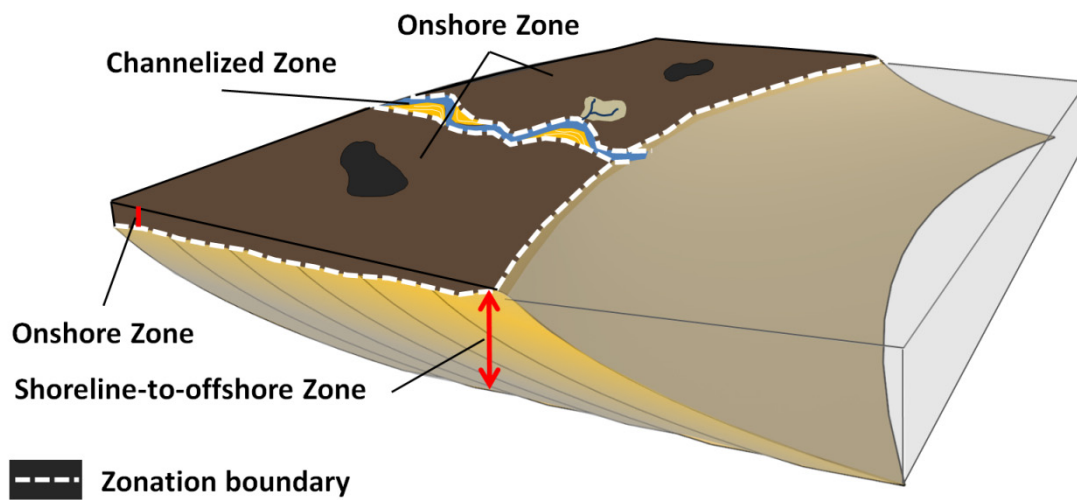


Figure 9. Differentiation between Shoreline-to-offshore, Channelized and Onshore Zones in a marginal marine setting. Each Zone in this case will be represented by an Element Complex or several Element Complexes, which belong to the zone (see text for discussion). See Table 2 for Zone definition.

Table 2). Table 1 lists all named Element Complex categories. Visual representations of these are shown in Figure 10.

Figure 6c illustrates how Element Complexes are defined in the Frontera system. Different Element Complexes in this case are defined in the wave-dominated portions of the system, not directly associated with the building of delta lobes (yellow), and the ones that are associated with delta lobes (orange and green). The lobes themselves are associated with two types of Element Complexes: Element Complexes directly linked to fluvial feeder systems that would internally be built by Channel Fill Elements and Mouth Bar Elements (green), and Element Complexes that are still clearly part of a lobe but are not directly associated with a distributary channel (orange). It must be noted that the orange and yellow Element Complexes in this case are wave-dominated and are both built by Beach Ridge and Swale Elements. Nonetheless, the geometries and orientations of the two are significantly different. Additionally, the orange Element Complexes are directly associated with distributary channels and the yellow Element Complexes are not. Thus they are placed in different Element Complex depositional process categories.

The Frontera Element Complexes can be named using the list of Element Complex categories shown in Figure 10 (see Figure 6d). By convention all Element Complex names contain a process definition followed by an Element Complex type: (i) Shoreline-to-offshore Zone types: Mouth Bar, Lobe, Beach, Tidal Flat, Barrier and Lagoon Element Complexes; (ii) Channelized Zone types: Channel and Funnel Basin Element Complexes; (iii) Onshore Zone types: Onshore Element Complex. The yellow Element Complexes in Figure 6 are therefore labeled as W-Beach Element Complexes. The green Element Complexes are labeled as 'Wf-Mouth Bar Element Complexes,' and the orange Element Complexes are labeled as 'Wf-Lobe Element Complexes.' The reason for the Wf process classification is the proximity to a river mouth and the assumption that the Element Complex will be at least partially fluvial influenced (e.g., due to presence of along shore fluvial-derived sand and mud transport and the influence of freshwater influx on the faunal populations and behaviors).

Two aspects of Element Complexes must be addressed. The first one is that Element Complexes represent bodies of predictable facies variability, which is related to the distribution of Elements internal to the units. Individual Element Complexes with very different facies characteristics can, for example, form as part of the up-drift side and the down-drift side of a single asymmetric delta lobe (Bhattacharya and Giosan, 2003). This is the reason why we suggest that vertical facies successions observed in core or outcrop will most easily be associated with individual Element Complex categories, which can be described and named, rather than entire depositional systems. The second aspect is that the same Element

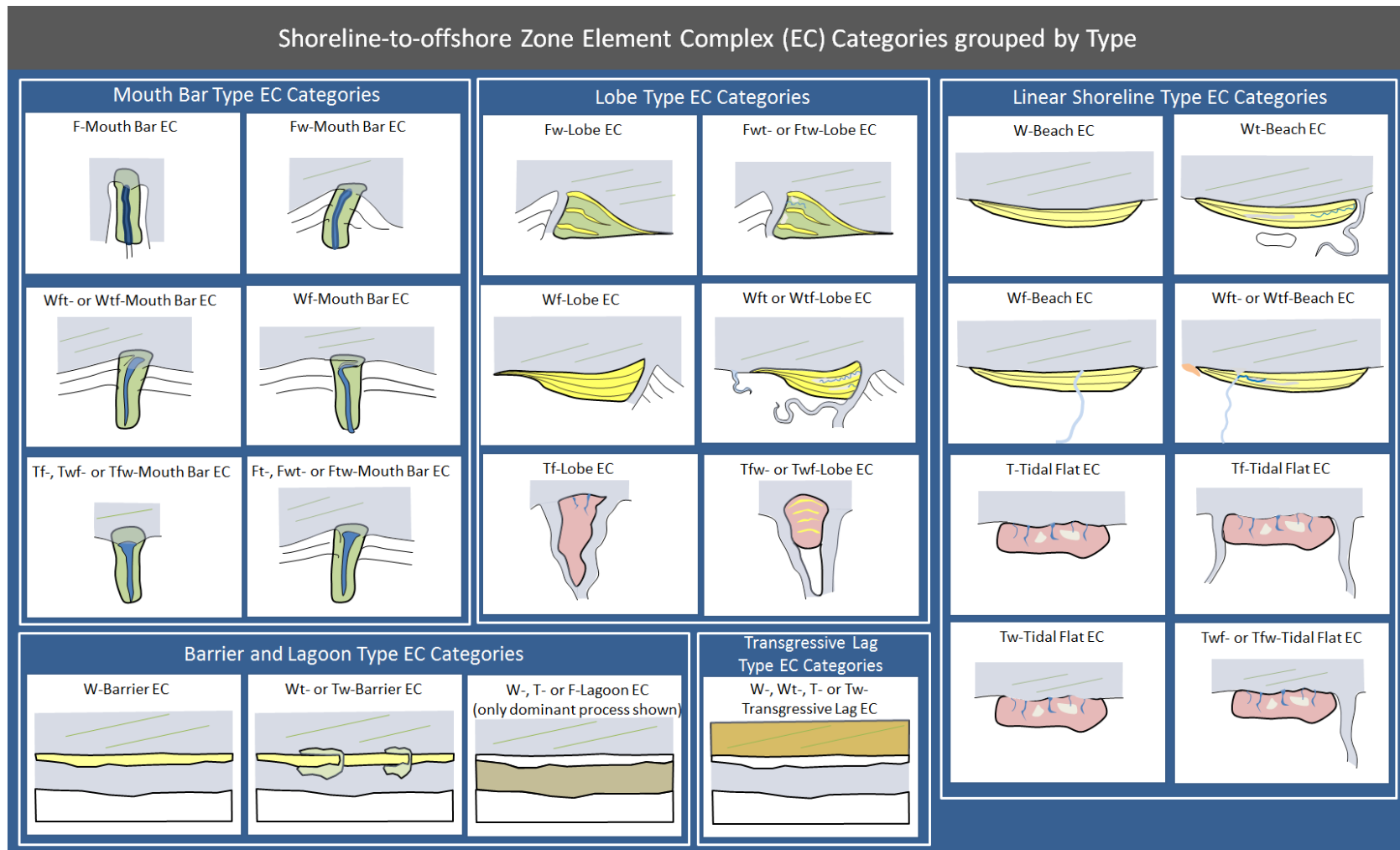


Figure 10. Illustrations of the named Element Complex (EC) categories in the architectural classification. Each Element Complex is shown in color, whereas adjacent Element Complexes are shown as outlines only. Element Complexes are subdivided into three Zones: Shoreline-to-offshore, Channelized, and Onshore (see Figure 9 and Table 2 for definition of Zone). Each Element Complex is named by assigning it a process category (e.g., Wf) and an Element Complex type (e.g., Lobe). Shoreline-to-offshore Zone Element Complexes can be of 'Lobe,' 'Beach,' 'Tidal Flat,' 'Barrier,' and 'Lagoon' types. Channelized Zone Element Complexes can be of 'Channel' and 'Funnel Basin' types. All onshore Element Complexes are assigned an 'Onshore' type. Element Complexes are the architectural units most easily related to observations in individual core or outcrop sedimentary logs. A list of Element Complexes and subdivision of these categories into types is provided in Table 1. Definition for the term Element Complex is provided in Table 2.

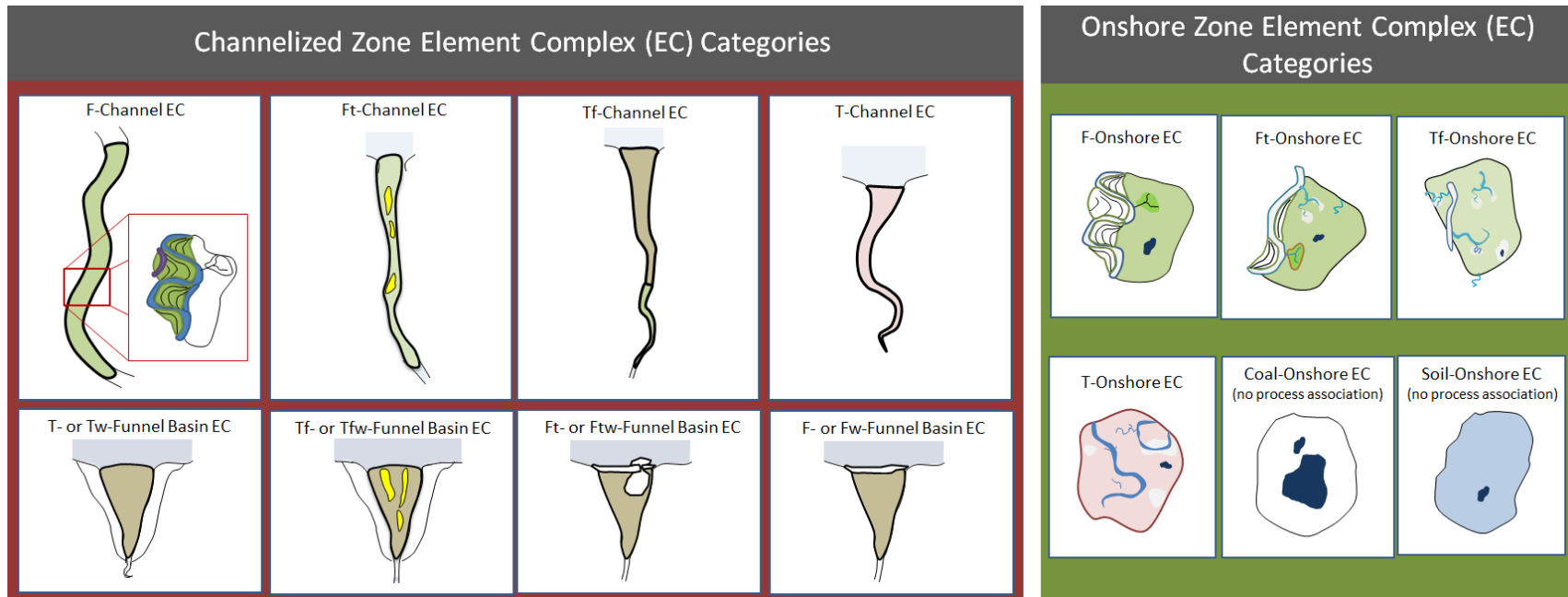


Figure 10 (Continued).

can occur in more than one Element Complex type (e.g., Mouth Bar Elements occur in all Mouth Bar Element Complex types listed in Figure 10). The geometry and internal facies properties of such Elements are likely to be different in response to the process regime controlling their formation.

The other architectural unit which is part of Hierarchy Level II is the Element Complex Set (ECS). It represents a grouping of genetically related Element Complexes affected by the same regional shoreline process conditions. The Element Complex Set is discussed in the following section due to its dependence on the definition of a parent Element Complex Assemblage beforehand (see next section for further explanation).

4.1.3 Hierarchy Level III- Element Complex Assemblage (ECA)

Genetically related Element Complexes can be grouped into larger-scale architectural units; Element Complex Sets and Element Complex Assemblages, which are defined by bounding surfaces marking either (i) major geospatial reorganization of a shoreline or (ii) significant process change on a shoreline. The difference between an Element Complex Set and an Element Complex Assemblage is related to the process classification of the unit relative to adjacent older and younger units. A group of Element Complexes is defined as an Element Complex Set if no process category change occurs across its external bounding surfaces, whereas it forms an Element Complex Assemblage if such a process change can be demonstrated. Element Complex Assemblages are placed in Hierarchy Level III of the architectural classification, while Element Complex Sets are placed in Hierarchy Level II together with the Element Complex architectural unit since both the EC and ECS units can share a parent-child relationship with the same Element Complex Assemblage category as parent (Figure 5). A list of Element Complex Assemblage categories is shown in Figure 11 and Table 1. A list of Element Complex Set categories is shown in Table 1.

Figure 12 clarifies how and when Element Complex Sets and Element Complex Assemblages are defined. The most straightforward example of an Element Complex Set in a regressive deltaic system is a pulse of shoreline progradation after an avulsion of a trunk distributary channel. Figure 12a illustrates such a scenario, where a fluvial-dominated deltaic system builds a depositional unit on the right, which becomes abandoned after channel avulsion (abandoned lobe), and then builds a depositional unit on the left (active lobe). Both depositional units in this case will form separate Element Complex Sets (F-Element Complex Sets: dashed red line) and the two units together will form a single Element Complex Assemblage (F-Element Complex Assemblage: solid red line). The reason why both Element Complex Sets will form a single Element Complex Assemblage has to do with the fact that they share the same process category. Not all systems will require the definition of Element Complex Sets. Figure 12b shows an example where

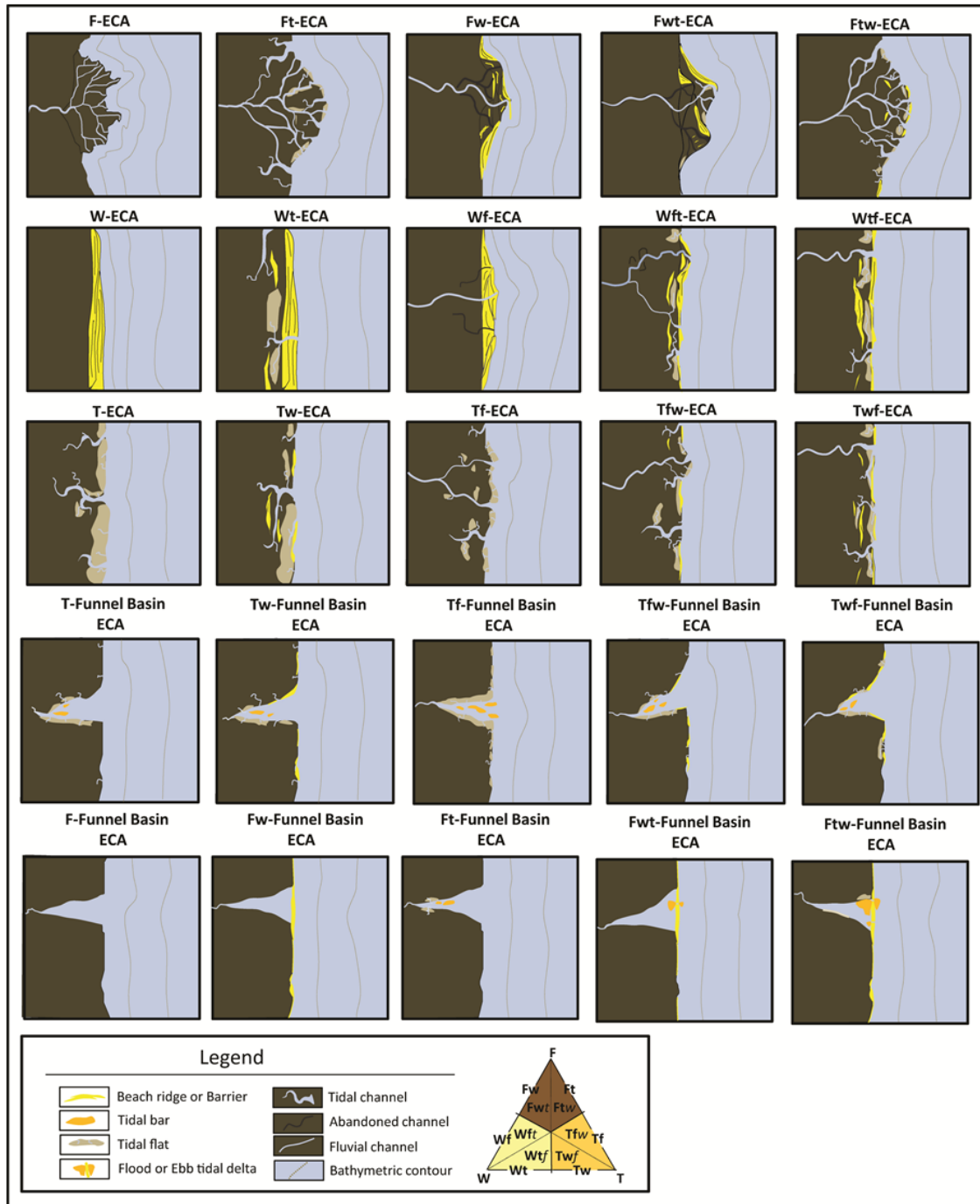


Figure 11. A List of Element Complex Assemblage (ECA) categories with example illustrations (modified from Ainsworth et al., 2011). Element Complex Assemblages are always associated with one of the fifteen process categories shown in Figure 1 and can be thought of as the regional shoreline response to a range of wave, tide and fluvial conditions. Each Element Complex Assemblage category can be constructed by a limited number of Element Complex categories (Figure 10, Table 1), with the link between the two represented by parent-child relationships. See Table 1 for list of Element Complex Assemblage categories and Table 2 for Element Complex Assemblage definition.

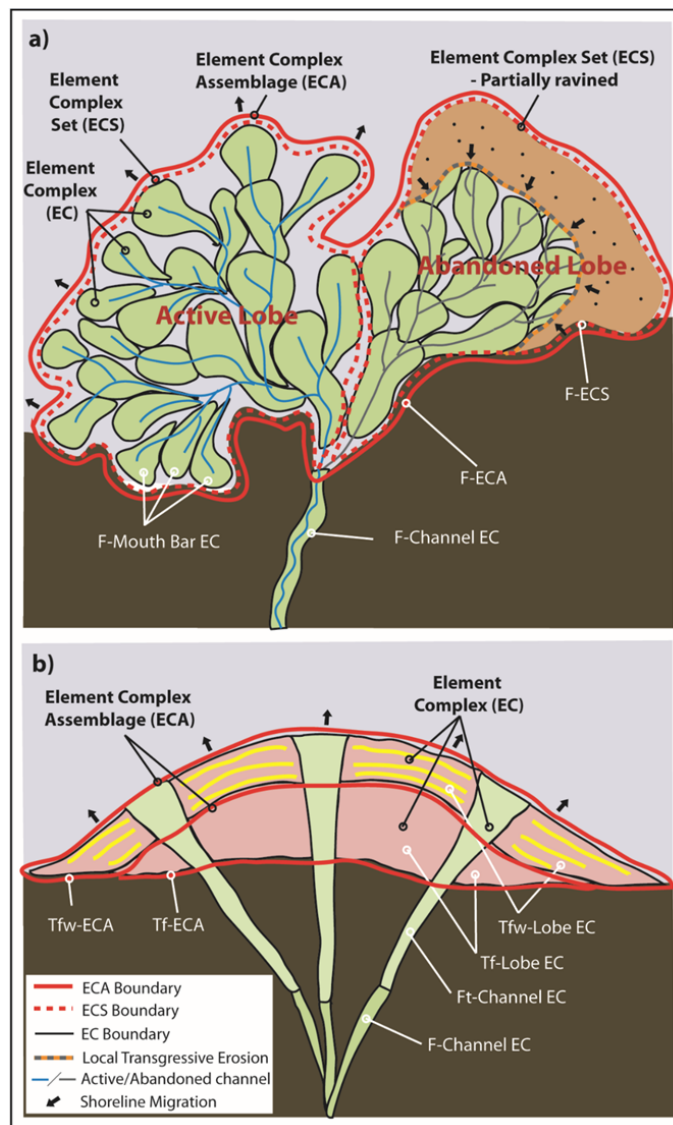


Figure 12. Examples of how different architectural terms can be applied to systems using plan-view geometries. (a) An example of a fluvial-dominated (F) system that can be described by different architectural units and categories. The system has been built by two pulses of progradation separated by an avulsion episode. The deposits from the first pulse (abandoned lobe) are being partially ravined, while the deposits of the second pulse (active lobe) are currently prograding seaward. Black arrows indicate direction of migration of shorelines. Each lobe is internally represented by a number of Element Complexes related to individual distributary channel stretches (F-Mouth Bar Element Complex category). Each of the two lobes is defined as a separate Element Complex Set (F-Element Complex Set category), while both lobes together form a single Element Complex Assemblage (F-Element Complex Assemblage category). The reason for the two Element Complex Sets forming a single Element Complex Assemblage has to do with the same process classification category being shared by the two sets (see text for further explanation). (b) Definition of architectural units and categories in a mixed-influence system which has prograded unaffected by avulsion. The system can be characterized by a number of Lobe and Channel Element Complexes. The system can be subdivided into two Element Complex Assemblages (red lines). The first Element Complex Assemblage will be classified as a Tf-Element Complex Assemblage and the second one as a Tfw-Element Complex Assemblage due to the increased wave influence in the seaward portion of the system. Note that definitions of Element Complex Sets are not required as no major reorganizations of shorelines have occurred internal to both Element Complex Assemblages.

only Element Complex Assemblages need to be defined. In this case, a process change occurs in down-dip depositional direction during progradation due to an increase in wave influence through time. The change in process requires the definition of two separate Element Complex Assemblages (Tf-Element Complex Assemblage and Tfw-Element Complex Assemblage). Process categories in this case are determined by percentage area of internal Element Complexes. An Element Complex Set definition is not required since a process change is observed across the stratigraphic discontinuity.

The Element Complex Set and Element Complex Assemblage concept can be applied to the Frontera coastal deposits (Figure 6). Two pulses of shoreline progradation related to avulsion of distributary channels are suggested by the distribution of Element Complexes on the left-hand side of the system (Figure 6c, d). Each of these deposits is internally built by two Lobe Element Complexes and one Mouth Bar Element Complex. Since each pulse of progradation represents a major reorganization of the shoreline without an apparent change in process conditions (in both cases the process classification is Wf), both units form Element Complex Sets. The two Wf- Element Complex Sets collectively form an Element Complex Assemblage, which is labeled with a Wf-Element Complex Assemblage category. The three W-Beach Element Complexes on the right-hand side of the Frontera system are placed in a separate Element Complex Assemblage labeled as a W-Element Complex Assemblage. Hence definition of separate Element Complex Sets is not required. However, the definition of a separate Element Complex Assemblage is required as this interval belongs to a different process category than the adjacent Element Complex Assemblage which is a Wf.

4.1.4 Hierarchy Level IV – Element Complex Assemblage Set (ECAS)

Hierarchy Level IV of the architectural classification deals with deposition during regional pulses of shoreline migration during transgression or regression. Since in many cases such deposits will consist of more than one Element Complex Assemblage, we refer to such deposits as Element Complex Assemblage Sets. Element Complex Assemblage Sets can be of a regressive type (Regressive Element Complex Assemblage Set, RECAS) or a transgressive type (Transgressive Element Complex Assemblage Set, TECAS). The plan view and cross sectional expression of such deposits is illustrated in Figure 7. A Regressive Element Complex Assemblage Set is very comparable with the concept of a “parasequence” and the sequence stratigraphic nomenclature that follows this notion (see Neal and Ardeu, 2009), and it is directly comparable with the Regressive Systems Tract as defined by Embry and Johannessen (1992). A Transgressive Element Complex Assemblage Set is directly comparable to the Transgressive Systems Tract as defined by Embry and Johannessen (1992). The Frontera system (Fig. 6) would represent a

Regressive Element Complex Assemblage Set. The landward extent of the unit cannot be determined based on available information since it is overlain by recent alluvium (Psuty, 1965).

4.1.5 Hierarchy Level V – Transgressive-Regressive Sequence (T-R Sequence)

A Transgressive Element Complex Assemblage Set overlain by a Regressive Element Complex Assemblage Set is equivalent to a T-R Sequence as defined by Embry and Johannessen (1992). If the RECAS in this case is associated with subaerial erosion, such a deposit forms a depositional sequence. The T-R Sequence represents Hierarchy Level V of the architectural classification and provides the link between the classification and existing sequence stratigraphic nomenclature.

4.2 Application of the Architectural Classification to Different Data Sets and Scales of Observation

The architectural classification is designed to be used with different data types at the appropriate for them scale. Figure 13 shows the recommended steps in using the classification when utilizing ancient subsurface, outcrop and modern data. Certain types of ancient data such as continuous outcrops or multiple, closely-spaced cores in the subsurface will permit the subdivision of a stratigraphic interval into all architectural units and hierarchy levels. Presence of only wireline logs or vertical seismic sections, in contrast, will allow the subdivision of the stratigraphy into at most two architectural units. The general recommended procedure for all data types is subdividing an interval into T-R Sequences, Transgressive Element Complex Assemblage Sets and Regressive Element Complex Assemblage Sets before attempting to define architectural units in the lower hierarchy levels. The only exception to this is the use of horizon slice seismic sections, where, if data quality allows, Element, Element Complex and Element Complex Assemblage units may potentially be directly discernible. In all other cases, definition of Element, Element Complex, Element Complex Set, and Element Complex Assemblage architectural units will require facies observation, which will therefore only be possible if core, outcrop or modern data is available.

4.2.1 Ancient Systems

We recommend slightly different procedures when applying the classification to (i) continuous outcrops and (ii) localized outcrops or subsurface cores (Tables 3). In both cases, the regional stratigraphic framework, in the form of definitions of T-R Sequences, as well as Transgressive Element Complex Assemblage Set and Regressive Element Complex Assemblage Set intervals, should be established. Facies variability should then be assessed, which will be used for definitions of Elements and Element Complexes. The difference in procedure is expressed by the approach taken for assigning Element Complex Sets and Element Complex Assemblages. In most cases, continuous outcrop will allow for

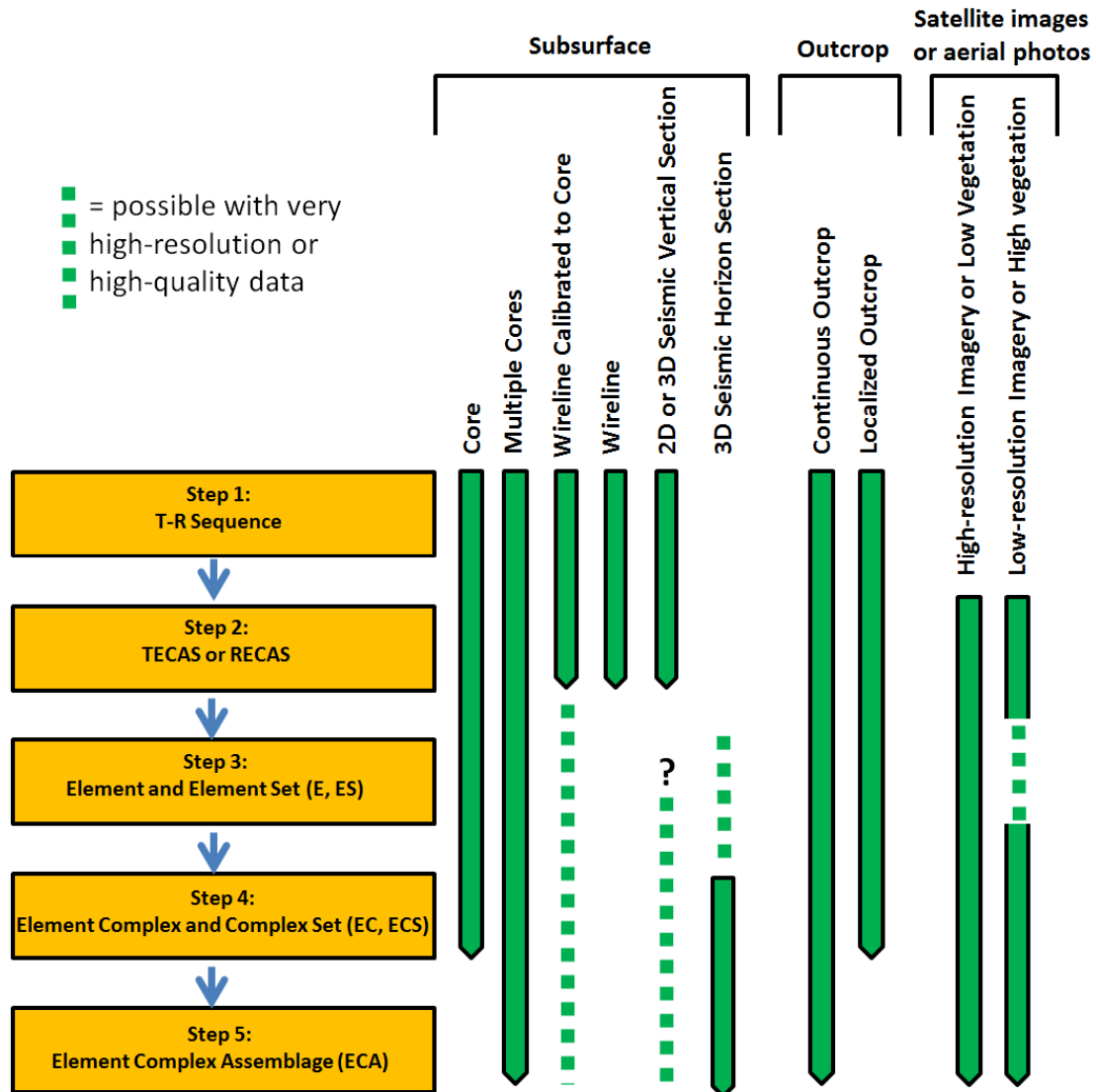


Figure 13. Use of the architectural classification with different types of subsurface-, outcrop- and modern data sets. The number of architectural units that can be determined in an area will be greatly influenced by the type of data available.

direct identification of Element Complex Sets and Element Complex Assemblages, based on the presence of discernible ‘bounding surfaces’ internal to Transgressive Element Complex Assemblage Set and Regressive Element Complex Assemblage Set intervals. Bounding surfaces can either be placed at locations showing marked change in process classification between adjacent units, as reflected by distribution of Elements and Element Complexes, or at locations where a major re-organization of the shoreline is suggested (e.g., evidence for minor flooding surfaces or minor erosional discontinuities). Once bounding surfaces are established, Element Complex Sets and Element Complex Assemblages can be directly defined (Figure 14). A unit will be labeled as an Element Complex Set if its process category remains the same relative to an adjacent (previously deposited) unit. A unit will be defined as an Element Complex Assemblage if a change in process category across the discontinuity can be demonstrated.

Core data and limited outcrop exposure are laterally discontinuous and do not allow for the direct mapping of Element Complex Sets and Element Complex Assemblages as is the case for continuous outcrops. The presence of such architectural units and their bounding surfaces can therefore only be inferred (Table 3). Since the Element Complex is the largest architectural unit that can be unequivocally determined from a single stratigraphic log (e.g., a single described core), it is the distribution of Element Complexes that will be most useful for determining the types and areal extents of Element Complex Sets and Element Complex Assemblages. In cases when multiple data points are available, which allows for a detailed Element Complex framework to be established, the type and areal extent of a number of Element Complex Assemblages may be determined. This is possible since each type of parent Element Complex Assemblage will be associated with a limited number of children Element Complex types (Figure 5). In more data limited scenarios, where insufficient Element Complexes are identified to directly map individual Element Complex Assemblage types and distributions, an uncertainty approach can be utilized, where a range of potential Element Complex Assemblages based on the available data will be generated (see *Architectural Classification and Uncertainty* section for further discussion).

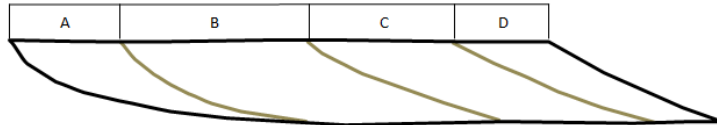
4.2.2 Modern Systems

A very different approach for mapping architectural units is suggested for modern systems where plan view geometries visible in aerial photographs and satellite imagery, rather than cross-sectional geometries or vertical sections, will be used. Modern systems described using such data will be heavily biased towards Regressive Element Complex Assemblage Set deposits, because a degree of shoreline progradation is required for architecture to become visible on aerial imagery. Observed architecture associated with modern Transgressive Element Complex Assemblage Set systems will, for the most part, be limited to active, present-day deposition at and behind the coastline, even though Elements, Element

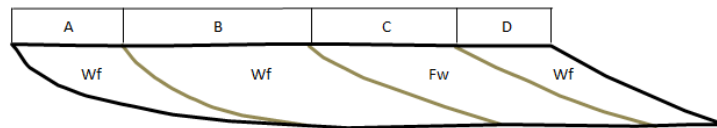
Step 1: Identify RECAS or TECAS interval



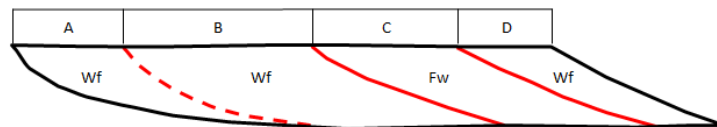
Step 2: Identify major bounding discontinuities and units



Step 3: Process classify each unit (based on E and EC)



Step 4: Draw ECA boundaries where neighboring process categories are different and ECS boundaries where neighboring process categories are the same



Step 5: Subdivide interval into ECS and ECA units

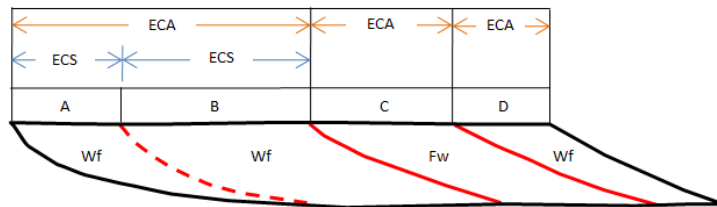


Figure 14. Methodology for assigning bounding surfaces and determining Element Complex Set (ECS) and Element Complex Assemblage (ECA) distributions using continuous outcrop.

Table 3: Recommended procedure for assigning different architectural units in ancient systems using various data sets such as continuous outcrop, core or discontinuous outcrop

<u>Transgressive-Regressive (T-R) Sequence - Continuous Outcrop and Core (or Discontinuous Outcrop)</u> 1) Subdivide the interval into maximum transgressive and maximum regressive surfaces. 2) Define T-R sequences (e.g., A, B, C, etc.) as maximum regressive surface bounded units.
<u>Element Complex Assemblage Set (ECAS) - Continuous Outcrop and Core (or Discontinuous Outcrop)</u> 1) Subdivide the interval into maximum transgressive and maximum regressive surfaces and define T-R Sequences. 2) Define Transgressive Element Complex Assemblage Set (TECAS) intervals as units bounded by maximum regressive surfaces at the base and maximum transgressive surfaces at the top. 3) Define Regressive Element Complex Assemblage Set (RECAS) intervals as units bounded by maximum transgressive surfaces at the base and maximum regressive surfaces at the top. 4) Label each TECAS and RECAS with name of parent T-R Sequence (e.g., TECAS A; RECAS A).
<u>Option I: Element Complex Set (ECS) and Element Complex Assemblage (ECA) (Continuous Outcrop)</u> 1) Within a single RECAS or TECAS category, mark major bounding surfaces across which a significant process change or shoreline reorganization can be demonstrated. 2) Determine the process classification for each new interval based on relative proportion of wave-, tide- and fluvial-dominated sedimentary structures and ichnology. 3) Identify each interval as an Element Complex Set or Element Complex Assemblage depending on whether process classification changes across its boundary. Identify intervals as Element Complex Sets if the process classification does not change and Element Complex Assemblages if the process classification changes (See Figure 14). 4) Name each Element Complex Set and Element Complex Assemblage based on its process classification (e.g., Fw-Element Complex Set; Wf-Element Complex Assemblage) <u>Option II: Element Complex Set (ECS) and Element Complex Assemblage (ECA) (Core and Discontinuous Outcrop)</u> 1) Only attempt to identify Element Complex Assemblages if several closely spaced cores or several outcrop sections are available in the same area. 2) Determine an Element Complex category for each present data point (e.g., Wf-Mouth Bar Element Complex) (see below) 3) If several adjacent Element Complexes are identified, consider if they were likely part of the same parent Element Complex Assemblage (i.e., if they co-occurred under a similar regional process regime at the shoreline), or are parts of more than one Element Complex Assemblage (if they did not co-occur under a similar regional process regime). 4) Determine process classification category (overall percentage of facies in internal Element Complexes) 5) Name the Element Complex Assemblage (e.g., Wf-Element Complex Assemblage) (see Figure 11)
<u>Element Complex (EC) (Continuous Outcrop and Core (or Discontinuous Outcrop))</u> 1) Within a single Element Complex Set or Element Complex Assemblage identify Shoreline-to-offshore, Channelized, and Onshore zones if such are present (Figure 9; see Table 2 for definition of Zonation). 2) Determine the process classification for each zone using proportion of sedimentary structures (or proportion of Elements that have already been assigned a dominant process). 3) For Shoreline-to-offshore zone intervals assign an Element Complex type: Mouth Bar, Lobe, Beach, Tidal Flat, Barrier, Lagoon (see Figure 10) 4) Categorize each Element Complex based on process classification and Element Complex type (Shoreline-to-offshore Zone: Mouth Bar, Lobe, Beach, Tidal Flat, Barrier, Lagoon Element Complex; Channelized Zone: Channel, Funnel Basin Element Complex; Onshore Zone: Onshore Element Complex) (e.g., Wf-Mouth Bar Element Complex) (see Table 1 and Figure 10).
<u>Element (E)</u> 1) Identify architectural Element categories based on lithofacies and ichnology.

Complexes, and Element Complex Assemblages in such systems can still be identified. Older architectural units that are part of the same Transgressive Element Complex Assemblage Set will be submerged below sea level and will therefore not be visible in aerial imagery. The architecture visible in modern systems will also be greatly affected by the resolution of available imagery (aerial photography or high resolution satellite imagery versus conventional satellite imagery) and the amount of vegetation cover in the area (in turn a function of climate, precipitation and anthropogenic influence) (Figure 13).

Modern systems are process-classified based on the presence of identifiable Elements and Element Complexes. Additional criteria such as the shape of the shoreline and the nature of channel networks can also be used as supplementary data (Vakarelov and Ainsworth, 2011). Table 4 outlines the recommended procedure for identifying different architectural units in modern systems. Identification of Element Complex Set and Element Complex Assemblage architecture shares similarities to the methodology outlined for continuous outcrops in the ancient, where key bounding surfaces marked by changes in depositional process regime or shoreline organization are identified. The definition of such surfaces in this case will be based on plan-view rather than cross-sectional geometries. As with continuous outcrops, Element Complex Sets will be defined when no change of process category occurs relative to an adjacent, previously deposited unit, whereas Element Complex Assemblages will be defined when such a change can be demonstrated.

Mapping Element Complexes in modern systems should be based on visible architecture coupled with considerations of depositional history. Element Complexes should not cross Element Complex Set and Element Complex Assemblage boundaries and should be defined based on (i) type and distribution of Elements within them, and/or (ii) presence of minor bounding surfaces across which changes in orientation of Elements of the same type can be demonstrated. Element Complexes are most easily defined if the depositional history of a system is well understood. This is why we recommend to couple Element Complex analysis with a paleogeographic reconstruction of a system, where pulses of shoreline progradation and position of fluvial channels through time are taken into consideration (Nanson et al., 2012a,b).

5. CASE STUDIES

The application of the architectural classification to a combined ancient outcrop and subsurface example and to a modern example is demonstrated below.

5.1 Application of the Architectural Classification to an Ancient System: Bearpaw-Horseshoe Canyon Formation Transition, Alberta, Canada

Table 4: Recommended procedure for assigning different architectural units in modern systems using aerial imagery

<u>Transgressive-Regressive (T-R) Sequence</u> - Not applicable in most Holocene systems when aerial imagery is used.
<u>Element Complex Assemblage Set (ECAS)</u> - Most Holocene successions show a single Regressive Element Complex Assemblage Set (RECAS) or Transgressive Element Complex Set (TECAS) in aerial imagery.
<u>Element Complex Set (ECS) and Element Complex Assemblage (ECA)</u> 1) Define major bounding surfaces (former shoreline positions) in depositional dip direction across which a significant process change or major shoreline reorganization can be demonstrated. 2) Define major bounding surfaces in depositional strike direction along which a major change in depositional environment or a major change in shoreline or paleoshoreline shape occurs. 3) Assign a process category to all surface-bounded intervals defined above, using the proportion of internal Elements belonging to a wave-, tide- and fluvial-dominated categories. This can be done by mapping and describing all Elements within an interval and measuring their relative areal extent proportions, or by visual estimate. 4) Compare the resultant process classification categories for all defined intervals. 5) If two or more adjacent intervals have been assigned the same process category, define these as Element Complex Sets and label them collectively as a single Element Complex Assemblage (see Figure 12) 6) If two or more adjacent intervals have been assigned different process categories, define these as separate Element Complex Assemblages. 7) Name the newly defined Element Complex Sets and Element Complex Assemblages (e.g., Wf-Element Complex Set; W-Element Complex Assemblage) (See Figure 11)
<u>Element Complex (EC)</u> 1) Define Element Complex Assemblages and Element Complex Sets (see above). 2) Define boundaries for Channel Element Complexes (channels and all associated deposits) 3) Map minor bounding surface that designate former shoreline positions across which there is an abrupt change in Element categories, proportion of Element categories or Element orientation. 4) Identify lobate portions of former shorelines. Separate them into a Mouth Bar Element Complexes and two adjacent Lobe Element Complexes if they contain a single distributary channel. If a lobe contains multiple distributary channels, define all inter-distributary areas as separate Lobe Element Complexes, and define all distributaries and their associated mouth bar deposits as Mouth Bar Element Complexes. 5) Identify funnel shaped portions of shorelines and define them as Funnel Basin Element Complexes. 6) Along straight stretches of shorelines, define Element Complexes based on differences in internal Element proportions and Element orientations. 7) Place all mapped Element Complexes in Shoreline-to-offshore, Channelized, or Onshore Element Complex types. 8) Assign a process category based on proportion of preserved fluvial, wave, and tide Elements for each identified Element Complex. 9) Name each Element Complex based on process category and Element Complex type (e.g., Wf-Mouth Bar Element Complex) (see Figure 10)
<u>Element (E)</u> 1) Identify mapable architectural Elements in aerial photographs or high resolution satellite imagery. 2) Assign process classification to the assigned Elements and determine their categories.

Excellent outcrop exposure of a mixed-influence, marginal marine system of the Campanian age (Upper Cretaceous) Bearpaw-Horseshoe Canyon Formation transition near Drumheller, Alberta, Canada allows for the testing and application of the process and architectural classification approach to an ancient system. A 60m-thick interval has been subdivided into six allostratigraphic units (A-F), each containing a transgressive and regressive phase (Ainsworth, 1991, 1992, 1994) (Figure 15). Units are bounded by coal horizons and contain various proportions of wave-, tide- and fluvial-influenced facies that can transition into one another over short lateral distances. Both Units C and E, for example, switch from a wave-dominated to tide-dominated mixed-influence systems over scales of kilometers. Large-scale channels of varying mudstone and sandstone fraction, containing tidal indicators are present in a number of the units, which further complicate and fragment the stratigraphy (Ainsworth, 1994). Shoreline trajectories are close to the horizontal suggesting that the entire succession was deposited under low-accommodation conditions.

The architectural classification is applied to an outcrop-based cross section tied to four nearby cored wells, which allows testing of the methodology using both outcrop and subsurface data (Figure 15). The section is 15km long and is oriented in depositional-dip direction. Mapping out of maximum regressive and maximum transgressive surfaces is a necessary first step when applying the architectural procedure (Table 3; Figure 16). This allows for definition of T-R Sequences, as well as Transgressive Element Complex Assemblage Set and Regressive Element Complex Assemblage Set subdivisions for each sequence (Figure 15a). All additional architectural units, such as Element Complex Assemblages and Element Complexes are assigned internal to a Transgressive Element Complex Assemblage Set or Regressive Element Complex Assemblage Set interval. To define these, we use conventional sedimentological techniques, where facies and facies associations are defined and their spatial distribution is analyzed. Lithofacies can be related to the Elements present in the system. Element Complexes should be defined first and their boundaries should be placed where major changes in facies associations occur in both a vertical and lateral sense (Table 3; Figure 15c). The facies distribution within each Element Complex then has to be analyzed in terms of relative distribution of depositional processes (fluvial, wave, and tide), and a process classification has to be assigned to the architectural unit (see Ainsworth, 2008; Ainsworth et al., 2011) (Figure 16). Determining the relative proportion of facies (i.e., Elements) can be done in a quantitative manner, where the proportion of each facies is measured and percentage facies is accurately determined, or it can be done in a qualitative manner where these are visually estimated. After the process classification for each Element Complex is determined, the Element Complex has to be named. Naming will depend on the process classification, the Zonation of the Element Complex (Shoreline-to-offshore, Channelized, or Onshore) (see Tables 2), and the Element Complex type within a

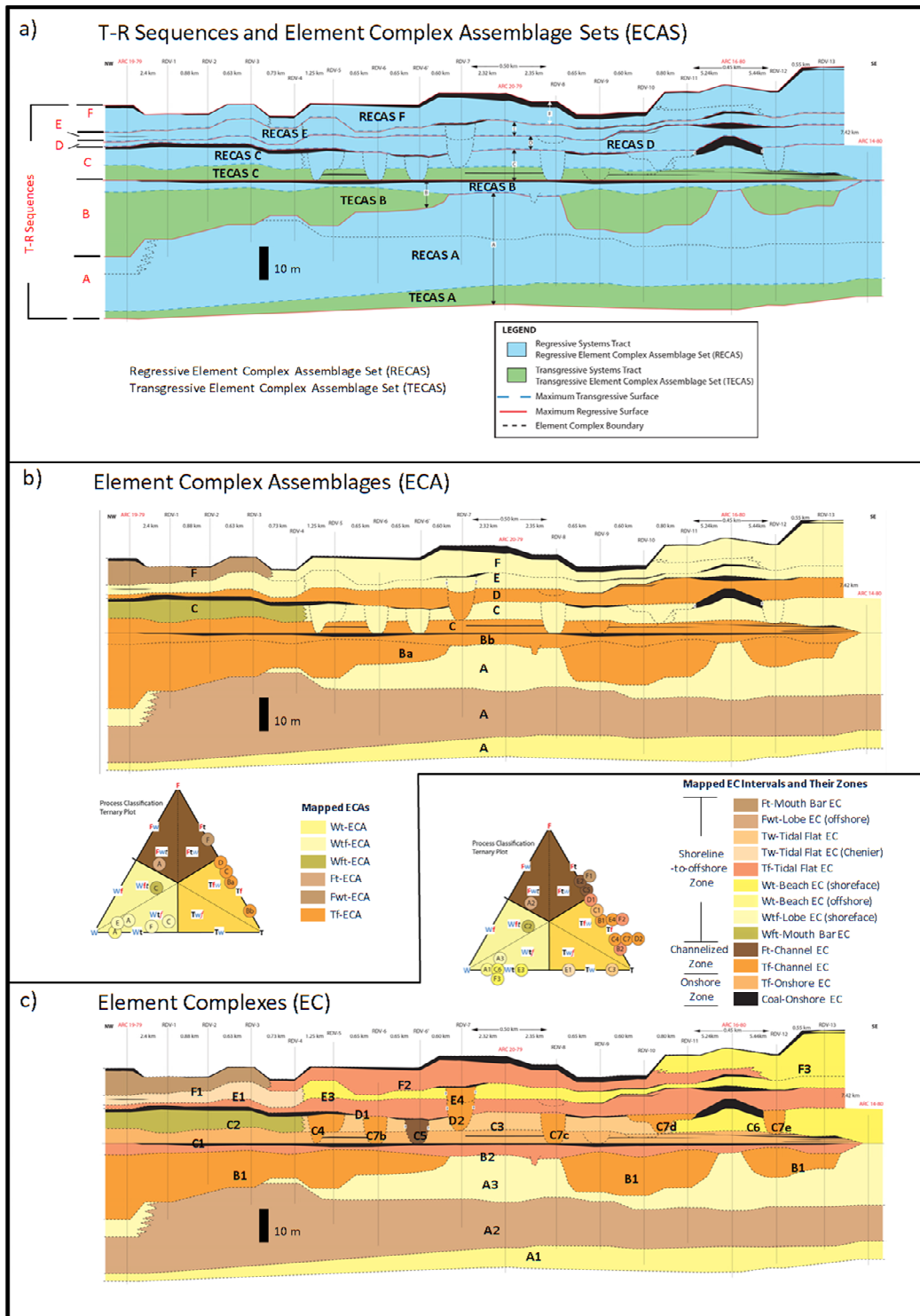


Figure 15. Application of the architectural classification to the Campanian-aged Bearpaw - Horseshoe Canyon Formation Transition near the town of Drumheller, Alberta, Canada. (a) Definition of maximum transgressive and maximum regressive surfaces, T-R Sequences, Transgressive Element Complex Assemblage Set (TECAS) and Regressive Element Complex Assemblage Set (RECAS) intervals. b) Definition of Element Complex Assemblage (ECA) intervals. c) Definition of Element Complex (EC) intervals (see text for further explanation).

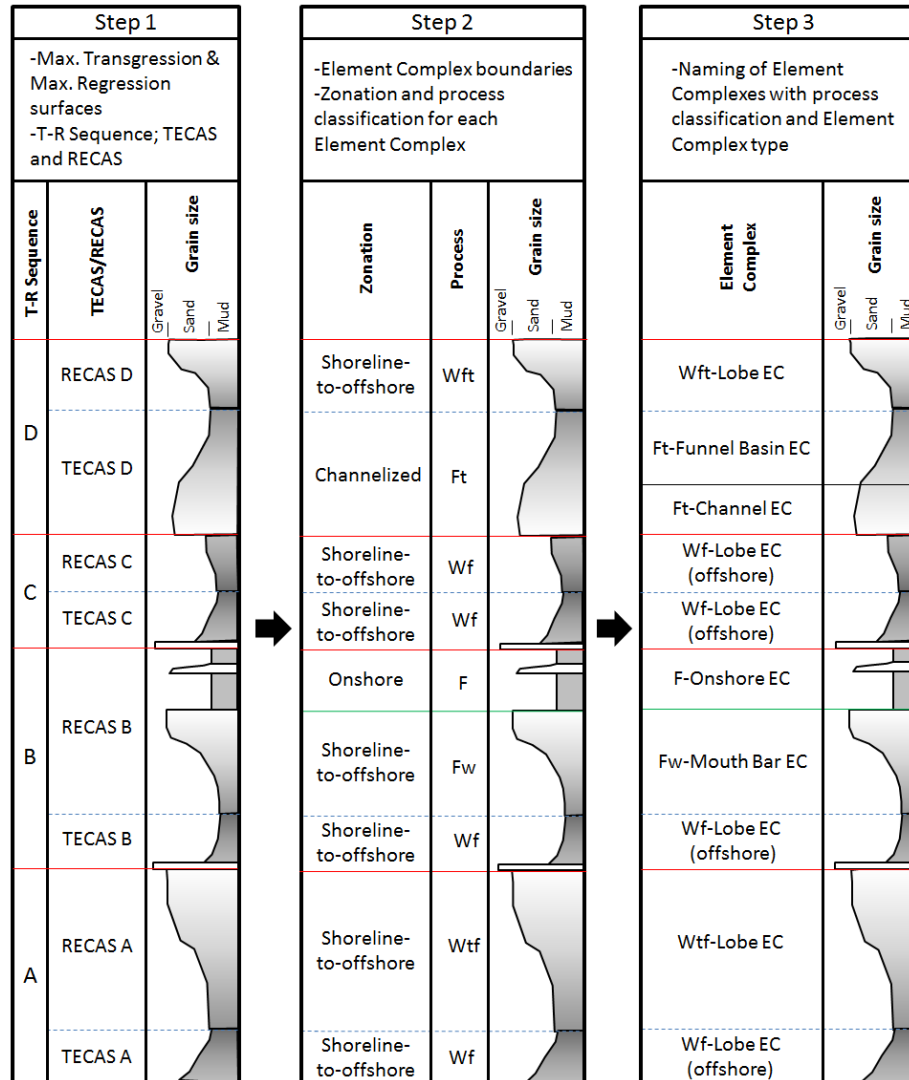


Figure 16. Methodology for defining architectural units in a single cored well in the subsurface (not based on real data). The grain size profile is for illustration purposes. No sedimentary structures are shown. Step 1: The interval is subdivided into transgressive and regressive packages by defining maximum regressive (red) and maximum transgressive (blue dashed) surfaces following conventional sequence stratigraphic methodology. This defines T-R Sequences, Transgressive Element Complex Assemblage Set (TECAS) and Regressive Element Complex Assemblage Set (RECAS) intervals. Step 2: Zonation (Shoreline-to-offshore, Channelized, or Onshore) is defined for each package. Additional key stratigraphic surfaces are marked (green lines) if more than one zone is present in a transgressive or regressive package. A process category is assigned to each interval based on percentage fluvial-, wave-, and tide-generated sedimentary structures. Step 3: Element Complex (EC) categories are defined and named for each zone depending on process category and Element Complex type (see Figure 10 and Table 1 for list Element Complex types). Note that the Channelized Zone in T-R Sequence D is subdivided into two separate Element Complexes (separated by black line) suggesting that the channelized zone likely represents an incised valley deposit. Direct definition of Element Complex Sets and Element Complex Assemblages has not been attempted based on the available data since such architectural units cannot be unequivocally determined from a single core. A range of applicable Element Complex Set and Element Complex Assemblage categories can be obtained for every determined Element Complex category using an uncertainty approach (see *Architectural Classification and Uncertainty* section in text).

zone (e.g., Wft-Mouth Bar Element Complex) (See Figures 10 and 16). Figure 15c shows the Element Complexes mapped in T-R Sequences A to G. Note the large number of Element Complexes assigned to Regressive Element Complex Assemblage Set C, where three Shoreline-to-offshore Zone Element Complexes and six Channelized Zone Element Complexes have been defined. This is necessary due to rapid down-dip changes in process domination of the shoreline-to-offshore facies and the large size of the channels which scale to the entire thickness of the unit. After definition of Element Complexes, Element Complex Assemblages can also be defined. Even though robust definition of Element Complex Assemblages requires understanding of the regional three-dimensional distribution of strata in the same Transgressive Element Complex Assemblage Set or Regressive Element Complex Assemblage Set, which is not available in a cross-sectional view, attempting to define Element Complex Assemblages is a useful exercise as it helps us understand the regional stratigraphic framework of the succession and the uncertainty associated with such an interpretation. Figure 15b shows our interpretation of Element Complex Assemblages internal to T-R Sequences A-G. In a manner similar to the naming of the Element Complexes, all Element Complex Assemblages are named with an associated process category (e.g., Wft-Element Complex Assemblage). No Element Complex Sets have been defined, since no bounding surfaces suggesting reorganization of the shoreline have been identified internal to any of the Element Complex Assemblages.

5.2 Application of the Architectural Classification to a Modern system: Mitchell delta, Northern Australia

The architectural classification was tested and applied to the modern Mitchell delta, North Queensland, Australia (Figs. 17 and 18). The system has prograded into the Gulf of Carpentaria during the Holocene under mixed-process conditions, resulting in a relatively complicated arrangement of architectural units, reflecting local variations in process domination through time. The entire system has prograded under low-accommodation conditions in water depths in the order of several meters (Nanson et al., 2012a). Fluvial influence in the system is suggested by the presence of distributary channels and their associated deposits. Wave influence is suggested by the presence of Beach Ridge and Chenier Elements that locally arrange themselves into Element Sets (Figure 17a, b). Tidal influence is suggested by the presence of tidal flats both at the shoreline and in low-lying areas behind the shoreline, presence of tidal channels and creeks, and presence of funnel-shaped fluvial channels at their mouths. Ainsworth et al. (2011) and Nanson et al. (2012b) showed figures where the Mitchell delta has been mapped at the Element level, where Elements have been placed into fluvial, wave, and tide categories. Using this approach, process domination could be determined for the entire system as well as for specific architectural units of the system (Figure 17b) (see Ainsworth et al., 2011). Using the element map and the aerial photography, we

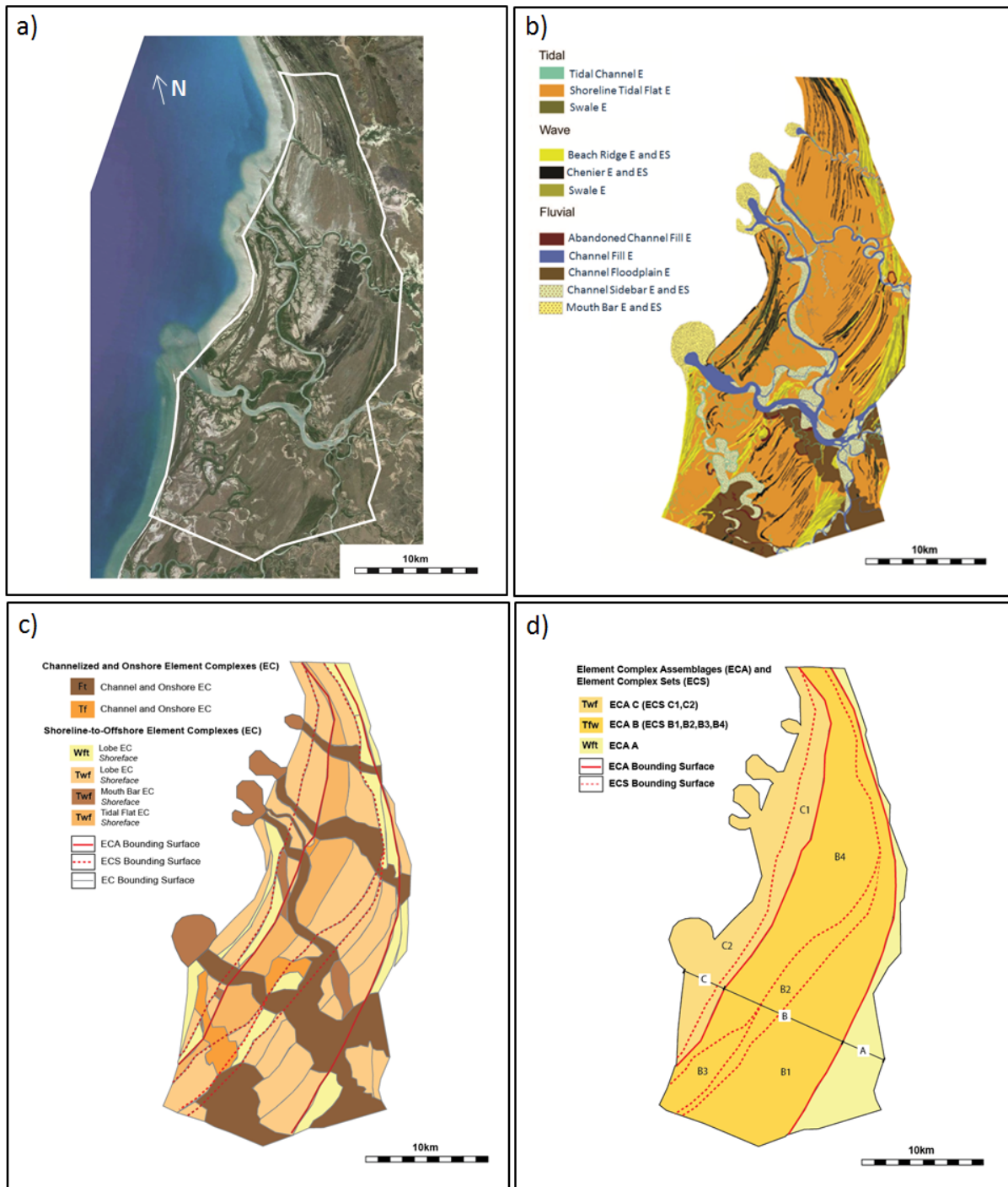


Figure 17. Application of the architectural classification to the modern Mitchell delta, Gulf of Carpentaria, Queensland, Australia. a) An aerial photograph of the delta with the study area outlined in white. b) Mapped Elements separated into fluvial, wave, and tide process category types (modified from Nanson et al, 2012b). c) Mapped Element Complexes, which have been defined based on distribution of internal Elements, fluvial channel locations and considerations of shoreline evolution. d) Mapped Element Complex Sets and Element Complex Assemblages (modified from Nanson et al., 2012a). Element Complex Sets are defined in cases when no change in process category occurs across their outer bounding surfaces, while Element Complex Assemblages are defined when such a change can be demonstrated. Element Complex Assemblage A, B and C are a part of a single Regressive Element Complex Assemblage Set formed during regional regression of the shorelines during the Holocene. See Figure 4 for a list of used architectural unit acronyms.

have defined and named larger-scale architectural units such as Element Complexes, Element Complex Sets and Element Complex Assemblages in the Mitchell system (Figure 17c).

The mapped portion of the Mitchell system represents a single Regressive Element Complex Assemblage Set. The RECAS can be subdivided into three Element Complex Assemblages (A, B and C) that represent regional changes in process classification during deposition (Figure 17d). Boundaries between the Element Complex Assemblages are considered to be significant bounding surfaces. Element Complex Assemblages B and C have further been subdivided into four and two Element Complex Sets, respectively, which mark major reorganizations of the shoreline not coupled with a change in process classification (Nanson et al., 2012a). Internal to each Element Complex Assemblage and Element Complex Set, a number of Element Complexes have also been defined (Figure 17c). The large number of Element Complexes reflects the highly mixed influence nature of the system, where relatively small shifts in process domination result in changes in local depositional conditions and the formation of different Element Complex types. This should be contrasted with the lesser number of Element Complex types in the Frontera system, discussed previously, where a simpler process regime results in less variability (Figure 6). Following the procedure outlined in Table 4, individual Element Complexes are defined by:

- (1) Identifying former shoreline position along which there is an apparent shift in process regime (minor bounding surfaces). For best results, this should be tied to a study of shoreline evolution through time.
- (2) Identifying channelized intervals and their associated deposits, defining Channel Element Complexes,
- (3) Deciding on process classification for each identified Element Complex and naming them.

The facies variability in the Mitchell Delta will closely mimic the Element Complex distribution outlined in Figure 17c. Due to the low accommodation setting, the surface expression of Element Complexes will also likely reflect the 3D architecture of the entire system, which is illustrated in the geocellular model of the system presented in Figure 18.

6. DISCUSSION

6.1 A Unified Modern and Ancient Classification Approach

The architectural classification seamlessly integrates modern and ancient observations and definitions of architectural units. This provides an opportunity to cross-reference observations from the two data types. Good quality outcrop-based ancient data, for example, provide the best available analogue information in terms of vertical cross-sectional view through stratigraphy (Figure 15). Modern systems, in contrast, contain abundant information on plan view geometries of architectural units (Figure 17). Identifying the same architectural units in modern and ancient systems can therefore provide information on the typical

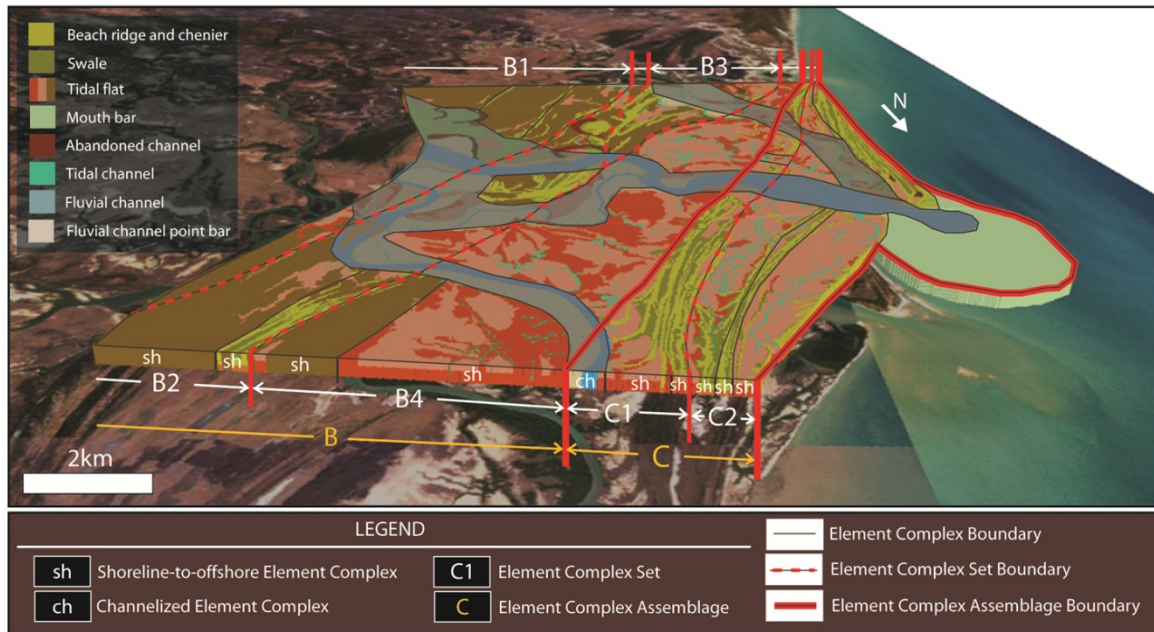


Figure 18. A 3D geocellular model of a portion of the Mitchell delta (Fig. 17) showing distributions of architectural units within one Regressive Element Complex Assemblage Set. The vertical section in the model is representative of the geometries normally observable in outcrop-based studies. The figure illustrates how the architectural classification can be readily applied to modern systems (plan view) and to ancient systems (vertical sections).

geometrical distribution of such units both in plan-view and cross-sectional sense. Such a combined approach will thus far better constrain the three-dimensional character of architectural units rather than using ancient or modern analogues in isolation. The vertical section through the 3D model shown in Figure 18 mimics the architecture typically visible in outcrop sections, graphically illustrating how the architectural classification can be applied to both ancient (vertical sections) and modern (plan-view) systems.

6.2 Parent-child and Sibling Relationships between Architectural Units

Defining architecture based on different hierarchical levels will be of little utility if links between the architectural units belonging to these levels cannot be established. Such links provide the logical framework between the different scales of observation within the same stratigraphic interval. The hierarchical nature of the architectural classification has been emphasized in various places in the text: Elements of the same type can form Element Sets; Elements and Element Sets can form Element Complexes; A group of genetically related Element Complexes formed under similar regional shoreline process conditions can form an Element Complex Assemblage, which in turn can be subdivided into separate Element Complex Sets; Element Complex Assemblages can further build Transgressive Element Complex Assemblage Set and Regressive Element Complex Assemblage Set deposits, which in turn form T-R Sequences (Figure 5). The way the architecture classification handles the links between these various architectural units is by defining parent-child relationships (Figure 19). By defining such relationships a nested architectural framework for any given deposit can be established.

Parent-child relationships are effectively represented as tree structures, where links can extend from hierarchy level to hierarchy level (Figure 5; Figure 19). All parent-child relationships extend across hierarchy level boundaries with the exception of Element to Element Set and Element Complex to Element Complex Set relationships which occur at an intra-hierarchy level (Figure 5). Figure 19 illustrates the links between the hierarchy levels, the architectural units that have been defined and the process classification. Note that T-R Sequence and Element Complex Assemblage Set (ECAS) units are not assigned a process classification while all other units are. All units are linked to one another through either parent-child or sibling relationships, represented with red and green arrows respectively. Sibling relationships refer to relationships between architectural unit categories which share the same parent architectural unit category.

Investigating the types of relationships that may exist for a given architectural unit category provides much useful information about the type of architectural likely to be encountered in the system. The number of specific Element Complex Assemblage-to-Element Complex category relationships will

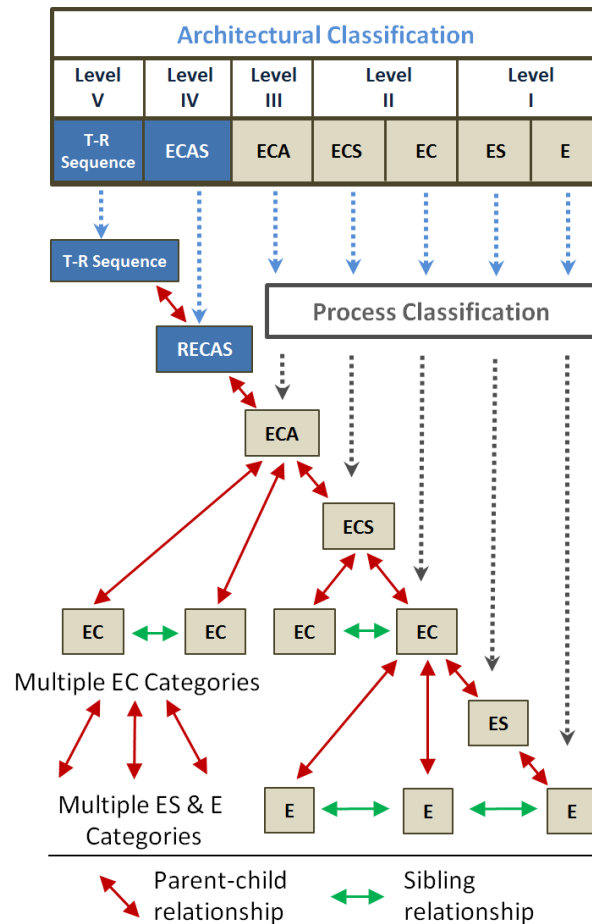


Figure 19. Architectural units and hierarchy levels that do and do not require process classification as part of their definition. Illustration of how parent-child and sibling relationships are defined between the different architectural units in the classification. Examples of parent-child relationships are also shown in Figure 5 and Table 1. See Figure 4 for a list of used architectural unit acronyms.

usually be limited, with many Element Complex Assemblage and Element Complex categories being mutually exclusive. A W-Element Complex Assemblage category, for example, will always have a parent-child relationship with a W-Beach Element Complex category but would never be associated with any of the Tidal Flat Element Complex categories, as the Element Complex Assemblage category in question is by definition not affected by tides. Element Complex categories which can share the same Element Complex Assemblage category as a parent will have sibling relationships to one another, indicating that such Element Complexes can co-exist and occur adjacent to one-another in the same parent architectural unit. Wf-Mouth Bar Element Complexes and Wf-Lobe Element Complexes have such a sibling relationship as they can both be children of a Wf-Element Complex Assemblage. Examining the relationships between architectural units belonging to different hierarchy levels provides a powerful toolset for uncertainty management in a marginal marine setting, and forms the subject of the next section.

6.3 Architectural Classification and Uncertainty

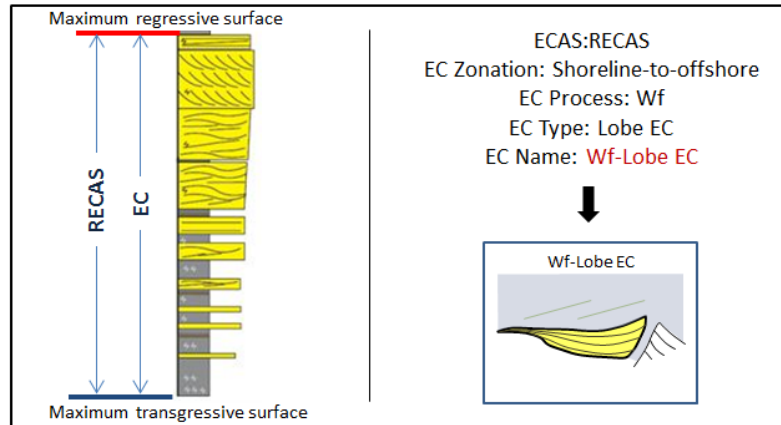
An important aspect of the architectural classification is the way it can be used for dealing with uncertainty in interpretation of marginal marine depositional environments. A common problem when working in the subsurface is having limited data on which to base depositional scenarios. Such scenarios are often used for building static geocellular models, which in turn can be used for building dynamic models and running reservoir simulations and are therefore of great practical importance. Non-expert interpretations of marginal marine intervals often end up being force-fitted into simplified fluvial-dominated, wave-dominated, or tide-dominated end-member categories. Identification of wave-generated facies in a single core, for example, commonly results in the development of a regional (linear) wave-dominated shoreline scenario; identification of mouth bars often results in a Mississippi-style fluvial dominated delta model. Since individual Element Complexes within mixed-influence systems can contain similar facies suites as Element Complexes in end-member systems, incorrectly considering only end-member depositional scenarios can lead to misleading assumptions about the architecture of a system.

A subsurface interpretation approach where observations from core are applied directly to the Element Complex unit, rather than to an overall depositional system, offers significant improvements in handling uncertainty. The Element Complex is the most suitable architectural unit to be associated with individual vertical stratigraphic logs from core or outcrop, and thus to facies observation. Since every Element Complex category has (i) a parent relationship with one or several Element Complex Assemblage categories (or Element Complex Set categories if such are defined), (ii) child relationships with a number of Element and Element Set categories, and (iii) sibling relationships with a number of other Element

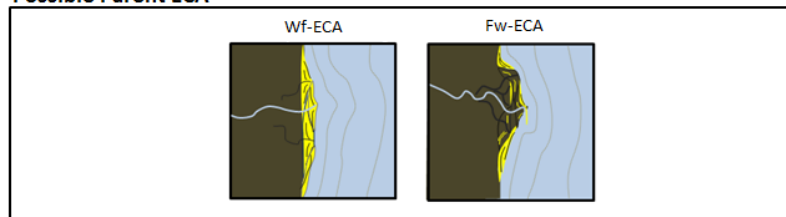
Complexes, identification of a single Element Complex can constrain the probability of encountering architectural units at the same or different hierarchy levels in the system (see Figure 19). Identification of EC₂ in Figure 5, for example, requires that ECS₁ should be defined as a parent and that ECA₂ should be defined as a grandparent to the unit. EC₂ also has child relationships with ES₁, E₂, and E₃ and a grandchild relationship with E₁. All of these architectural units can therefore be directly associated with the identified Element Complex. EC₂ has sibling relationships with EC₁, EC₃ and EC₄, indicating that the three types of Element Complexes can co-exist in the same system and occur laterally from EC₂. Even though these Element Complex siblings have not been directly observed in core, they can still be a component of the system and should be included in modeling scenarios. Since the same Element Complex category can have more than one Element Complex Assemblage category as a parent, which in turn can have other Element Complex categories as children, identification of an Element Complex will often result in a set of applicable architectural unit categories at all hierarchy levels. This would effectively capture the uncertainty of the system and suggest multiple depositional scenarios.

A simple example of such an uncertainty approach with actual architectural unit categories is illustrated in Figure 20. In this scenario, observations from a single core from a Regressive Element Complex Assemblage Set interval have determined the process classification to be Wf, the Zonation to be Shoreline-to-offshore, and the type of deposit to be a Lobe Element Complex. The deposit has therefore been identified as a Wf-Lobe Element Complex. Even though Wf-Lobe Element Complexes are an integral part of Wf-Element Complex Assemblages, they can also occur as part of Fw-Element Complex Assemblages. Since a single data point cannot differentiate between the two options, both Element Complex Assemblage types should be considered as possible parents to the identified Element Complex (Figure 20). Both Element Complex Assemblage categories can in turn have child relationships with other Element Complex categories, a list of which is also provided in Figure 20. These Element Complex categories have a sibling relationship to the identified Wf-Lobe Element Complex and can possibly co-occur in the same system. The Wf-Lobe Element Complex can therefore be laterally equivalent to five types of Shoreline-to-offshore Element Complexes, it can be overlain or incised by one type of Channel Element Complex, and can be overlain by three types of Onshore Element Complexes, if such deposits have been preserved (Figure 20). It must be noted that the example in Figure 20 starts with a single unequivocally identified Element Complex. Realistic subsurface scenarios often result in uncertainty in interpretation of Element Complex categories from core, where more than one Element Complex category fits observations (e.g., Wf-Lobe Element Complex vs. Wf-Beach Element Complex), which will further increase the number of applicable parent and sibling relations and the corresponding uncertainty. Additional data, in contrast, will often tend to decrease such uncertainty. A higher number of parent-child

Identification of EC unit



Possible Parent ECA



Possible Sibling EC

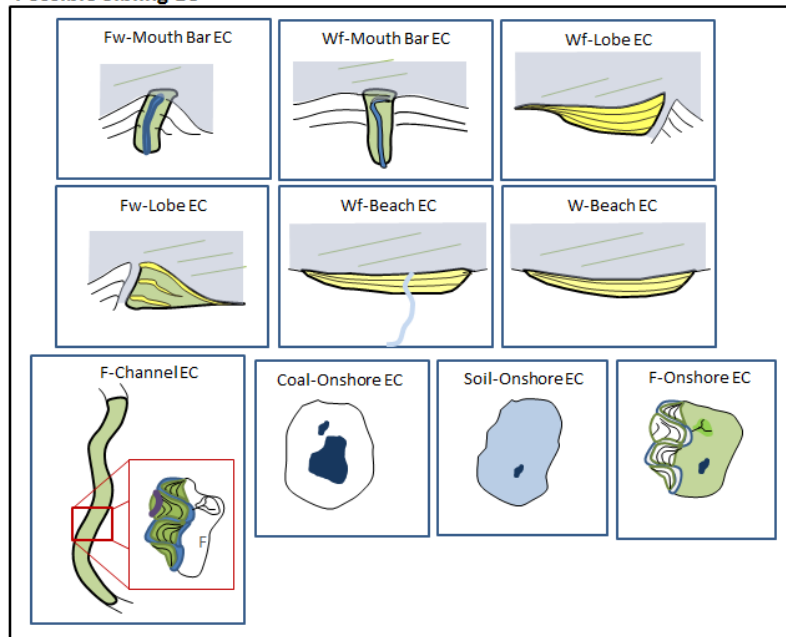


Figure 20. An example showing how the architectural classification can be used for managing depositional uncertainty in the subsurface. Top: A single cored well has been interpreted as a Wf-Lobe Element Complex (Wf-Lobe EC). Middle: A list of applicable parent Element Complex Assemblage (ECA) categories which can have a Wf-Lobe Element Complex as a child. Bottom: A list of all possible Element Complex (EC) categories that have child relationships with the identified Element Complex Assemblage categories above, including the Wf-Lobe Element Complex already identified. All Element Complex categories in the list have sibling relationships with the Wf-Lobe Element Complex and can be co-occurring in the same system.

relationships that need to be considered exist in systems affected by three rather than two processes. Analysis of such systems can be greatly simplified if tools such as decision trees or appropriate decision-making software are utilized (Vakarelov et al., 2010; Ainsworth et al., 2011). Properly designed database software, for example, can seamlessly analyze multiple parent-child relationships and provide lists of applicable parent, child, and sibling architectural categories to an identified architectural unit, efficiently handling the full range of uncertainty in the system.

6.4 The Effect of Rates of Accommodation Change on Regressive and Transgressive Element Complex Assemblage Sets

The nature of Regressive Element Complex Assemblage Sets and Transgressive Element Complex Assemblage Sets within a T-R Sequence can vary greatly depending on the rate of accommodation development during deposition. The accommodation settings can in turn be related to the shoreline trajectories during times of regression and transgression. Shoreline systems that are deposited by shoreline trajectories close to the horizontal can be said to be formed in stable to relatively low rates of accommodation creation conditions. This can occur under forced regressive, normal regressive and transgressive conditions. Element Complexes and Element Complex Assemblages that are deposited under normal regressive and transgressive conditions with positive shoreline trajectories significantly removed from the horizontal can be said to be deposited under relatively high rates of accommodation creation conditions. High accommodation conditions cannot occur during times of forced regression. A given accommodation setting can affect an entire Transgressive Element Complex Assemblage Set or Regressive Element Complex Assemblage Set unit, or, alternatively, such conditions can change internal to a single regressive or transgressive interval.

The type of stratigraphic architecture preserved will be different during times of low and high rates of accommodation development. Low accommodation settings will be characterized by Regressive Element Complex Assemblage Sets built by horizontal offlap of Element Complex Assemblages (and their internal Element Complexes) and Transgressive Element Complex Assemblage Sets which are thin or often absent, removed by wave and tide ravinement (Units C to F in Figure 15a; Figs. 17 and 18). The non-marine portions of such systems will usually be characterized by Channel Element Complexes that may truncate underlying Shoreline-to-offshore Element Complexes, and Onshore Element Complexes that are thin or entirely removed by ravinement. Accumulation of thick non-marine intervals under such conditions will be unusual. High accommodation settings, on the other hand, associated with Regressive Element Complex Assemblage Set deposition, will often result in a component of vertical stacking of both Shoreline-to-offshore and Channel and/or Onshore Element Complexes, with significant

preservation of non-marine strata (Figure 21). High accommodation Transgressive Element Complex Assemblage Sets will usually be depositional and will attain greater thickness than their low accommodation counterparts.

6.5 Stacking of Element Complexes and Element Complex Assemblages during Regression and Transgression

Element Complexes are defined as belonging to three different Zones (Shoreline-to-offshore, Channelized, and Onshore) (Figure 9). As systems evolve, especially in high accommodation settings during normal regression and transgression, stacking of sibling Element Complexes usually occurs. A normal regressive system will, for example, form a vertical succession consisting of a Shoreline-to-offshore Element Complex, overlain by Onshore and/or Channel Element Complexes as illustrated in Figure 21. The figure shows a conceptual scenario of a regressive fluvial-dominated, wave-influenced system (forming an Fw-Element Complex Assemblage) characterized by significant lateral shifting of sediment depocenters related to channel avulsion. The delta growth pulses in this case represent separate Element Complex Sets, which are formed as part of the same parent Element Complex Assemblage (see Figure 21). The time snapshots are shown on the right and an idealized vertical succession formed at the position of the red circle is shown on the left. The vertical succession is built by three stacked Element Complexes (each of which belongs to a different Element Complex Set) that are genetically related and form as part of the same parent Element Complex Assemblage. The bottom portion of the succession is represented by a Shoreline-to-offshore Element Complex (Fw-Lobe Element Complex). This Element Complex is overlain by a distributary channel (F-Channel Element Complex) related to a new Element Complex Set that is forming further seaward from the data point. The F-Channel Element Complex is in turn overlain by an F-Onshore Element Complex deposited at the time of the progradation of an even younger Element Complex Set. The evolution of the system therefore results in the vertical stacking of three different Element Complexes.

Three-dimensional stacking of Element Complexes also occurs in transgressive systems. Estuarine incised valley fills (Boyd et al., 1992; Dalrymple et al., 1992) can be represented by a range of Channelized Zone Element Complex types (F, Ft-Channel Element Complexes), Funnel Basin Element Complex types (F, Ft, Tf-Funnel Basin Element Complexes) and Barrier Element Complex types (W, Wt, Tw-Barrier Element Complex) that transition laterally into one another, as part of a parent Funnel Basin Element Complex Assemblage (Figure 11). The exact Element Complexes present will depend on the processes affecting the system, which as in the case of regressive systems can often be mixed-influence in nature.

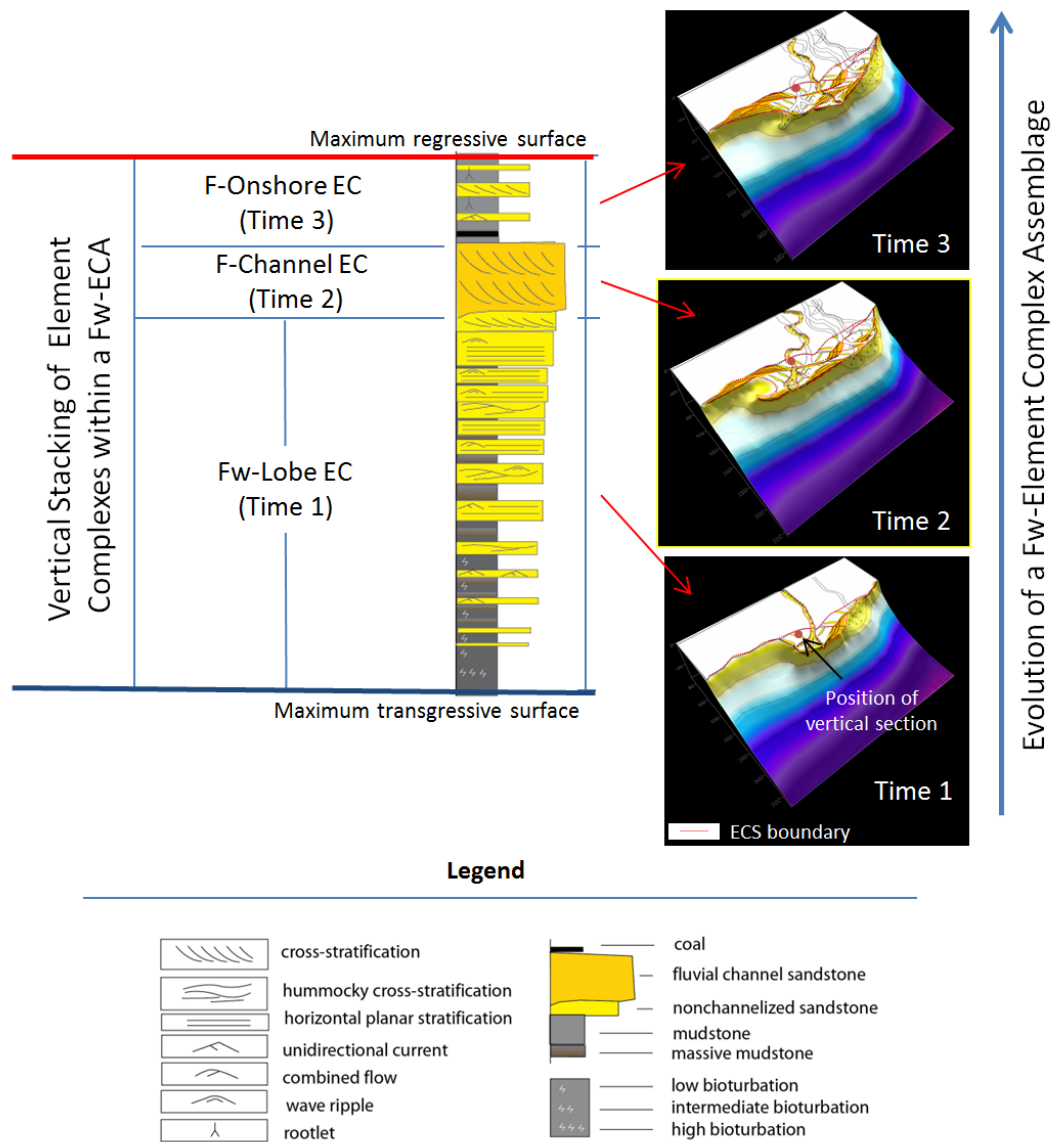


Figure 21. A conceptual figure showing the vertical stacking of Element Complexes to form an Element Complex Assemblage during normal regression. A fluvial-dominated, wave-influenced system (Fw-ECA), affected by frequent river avulsion and internally built by a number of Element Complex Set (ECS) intervals, is shown at three periods during its evolution (Time 1, 2, and 3). The red circle marks the position of the vertical succession shown on the left-hand side. Time 1 shows the formation of an Fw-Lobe EC, while Times 2 and Time 3 show the formation of an F-Channel EC and an F-Onshore EC, respectively. Note that every Element Complex belongs to a different Element Complex Set, all of which are a part of the same Element Complex Assemblage. No scale intended.

Facies models for estuaries (see Dalrymple et al., 1992; Boyd et al., 2006) can be represented by vertically stacked Element Complexes that represent the regional transgressive backstepping of a shoreline (Transgressive Element Complex Assemblage Set, TECAS) overlain by a prograding regressive package (Regressive Element Complex Assemblage Set, RECAS). To illustrate how the architectural classification can be applied to estuarine systems, and transgressive systems in general, we have re-drawn the well-established wave-dominated estuary schematic section from Dalrymple et al. (1992; Figure 14) (Figure 22). The new nomenclature is shown in red letters and the original nomenclature is shown in black letters (Figure 22). Note that according to our definition, the term wave-dominated estuary is a misnomer, since only a small part of such a system (the Barrier Element Complex) is wave-dominated. We refer to barrier-fronted estuaries as Fw-, Fwt-, or Fwt-Funnel Basin Element Complex Assemblages (Figure 11), depending on the relative proportions of wave and tide facies. We consider such systems to be fluvial-dominated as fluvial processes (and not wave processes) are likely to dominate most of the area behind the barrier.

The schematic lithofacies logs in Figure 22 have been separated into transgressive and regressive portions, which set the boundaries of Element Complexes and Element Complex Assemblages internal to them. Individual vertical facies successions are first interpreted in terms of Element Complexes, with a number of different Element Complex types defined. Note that we have lumped the barrier and related flood tidal delta into a single Element Complex that will be designated with a Wt or Tw process category, depending on the relative proportions of the two processes. The relationships between genetically-related Element Complexes can then help us define Element Complex Assemblages. Separate Element Complex Assemblages will be defined in the TECAS and RECAS portions of the system. These will be assigned Fwt-Funnel Basin Element Complex Assemblage categories, based on the distribution of internal Element Complexes and their processes. If all identified Element Complexes in the TECAS or RECAS portion of the system are not siblings (i.e., cannot have a common parent), this would be an indication that more than one Element Complex Assemblage must be present in the system, suggesting that the process regime must have changed during deposition. Defining estuarine fills in terms of stacked and laterally transitioning three-dimensional Element Complexes easily accounts for and describes the architecture, the mixed influence nature, and complexity of such systems.

The estuarine discussion can be extended to simple- and compound-fill incised-valley deposits (see Boyd et al., 2006). Lateral and vertical stacking of Element Complexes can describe the fill of incised valleys regardless of whether they occur in the non-marine, marginal marine or the fully marine spectrum. Unlike single story channels and channel belts that are best regarded by the classification as individual Channel Element Complexes, incised valleys usually contain sets of vertically or laterally stacked Element

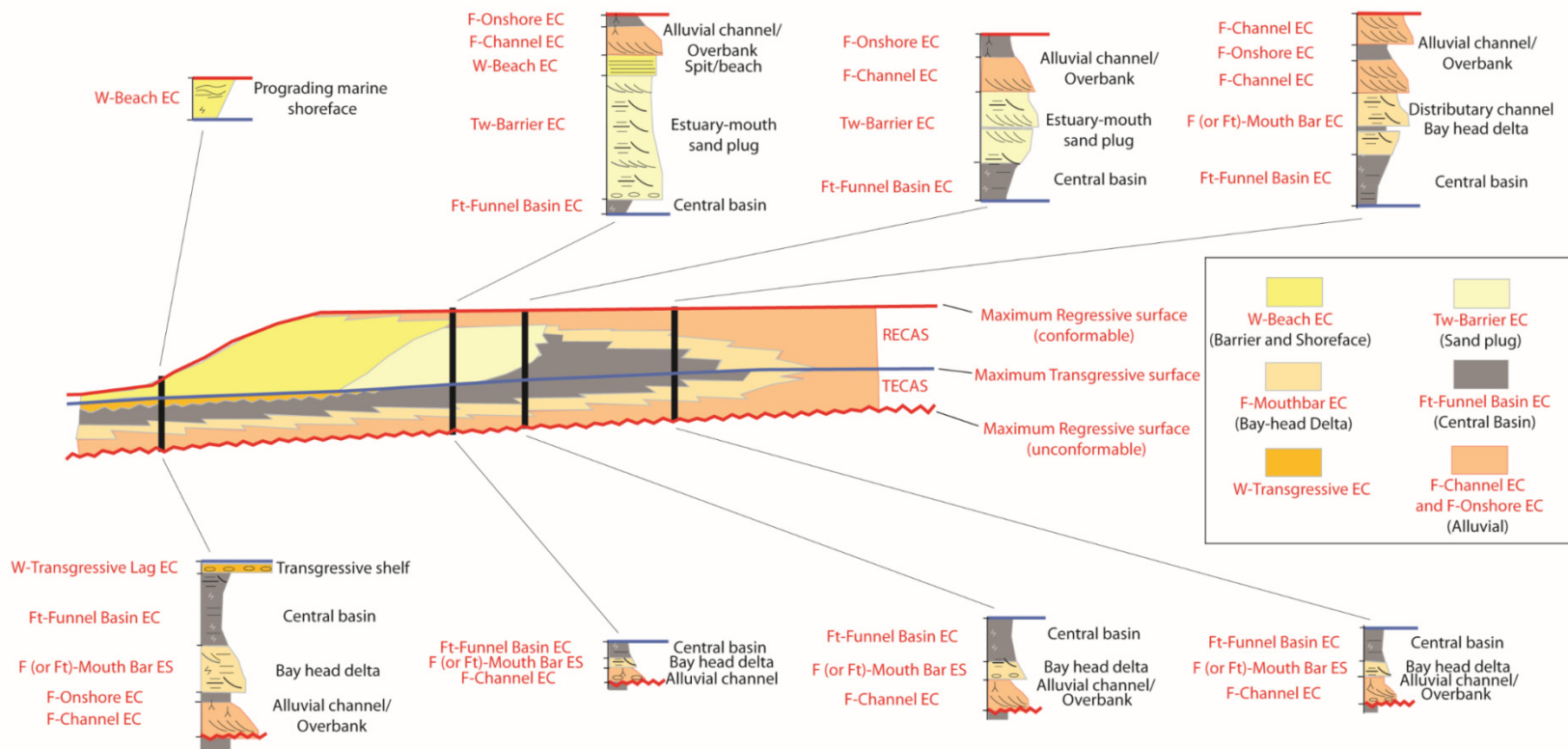


Figure 22. Application of the architectural classification to a well-established model for 'wave-dominated estuaries' from Dalrymple et al. (1992). The schematic logs from Dalrymple et al. (1992) have been redrawn and have been separated into transgressive (Transgressive Element Complex Assemblage Set, TECAS) and regressive (Regressive Element Complex Assemblage Set, RECAS) portions. The legend for the figure is shown in Figure 21. Note that separate Element Complex (EC) intervals have been defined in the transgressive and regressive portions of the system. The new process and architectural nomenclature is shown in red letters, while the original nomenclature is shown in black letters. All Element Complexes in the TECAS and RECAS portions of the system can also be assigned to two different Fwt-Funnel Basin Element Complex Assemblages.

Complexes that may belong to the same or to different parent Element Complex Assemblages. These Element Complexes can be of the same category (e.g., F-Channel Element Complex) or of different categories (various types of Channel and Onshore Element Complexes, Funnel Basin Element Complexes and Shoreline-to-offshore Element Complexes). The organization of stacked Element Complexes into higher level architectural units will depend on the relative sea-level history that has affected the deposit. In simple-fill incised valleys, Element Complexes and Element Complex Assemblages represented by the valley fill are the product of a single T-R Sequence, with varying proportion of Transgressive Element Complex Assemblage Sets and Regressive Element Complex Assemblage Sets represented. In compound incised valleys, affected by more complex sea-level history, the valley may be subdivided into a number of Transgressive Element Complex Assemblage Set and Regressive Element Complex Assemblage Set deposits and can be represented by more than one T-R Sequence. Applying the architectural scheme to estuarine and incised valley deposits represents an efficient way of describing the highly variable often mixed-influence nature of such deposits.

7. CONCLUSIONS

We present a new intra-parasequence scale architectural classification for marginal marine systems that is an extension of the process-based classification of Ainsworth et al. (2011). The classification addresses sub-parasequence scales of stratigraphy that often control fluid flow in these types of reservoirs. The architectural classification uses seven types of architectural units which are placed on five stratigraphic hierarchy levels that bridge the scale gap between sedimentology and sequence stratigraphy. Individual architectural units are linked to one another through parent-child and sibling relationships. The series of links form tree structures that span all hierarchy levels.

The newly defined architectural units and their corresponding levels are as follows: Hierarchy Level I – Element (E) and Element Set (ES); Hierarchy Level II – Element Complex (EC) and Element Complex Set (ECS); Hierarchy Level III – Element Complex Assemblage (ECA); Hierarchy Level IV – Element Complex Assemblage Set (ECAS) which can be transgressive (Transgressive Element Complex Assemblage Set, TECAS) and regressive (Regressive Element Complex Assemblage Set, RECAS); Hierarchy Level V – T-R Sequence. All architectural units are three-dimensional sediment bodies and can be characterized by their plan-view and cross-sectional geometries. All units except T-R Sequences and Element Complex Assemblage Sets are associated with one of fifteen depositional process categories.

The new architectural scheme is equally applicable to modern and ancient systems, which permits direct comparison of geometries of architectural units between the two data types

The proposed classification has important implications for dealing with uncertainty in the subsurface. Different data types, such as core, wireline data or seismic can be directly applied to the most appropriate hierarchy levels and architectural units. Lists of potential architectural unit categories belonging to other hierarchy levels can then be generated based on parent-child and sibling relationships. It is argued that observations in core are best applied to Element Complex architectural units, which represent components of depositional systems, rather than entire depositional systems. A single facies succession in core (i.e., an Element Complex) can then be related to one or several parent Element Complex Assemblage categories, allowing the inherent uncertainty in the interpretation to be effectively managed. The predictive power of the classification then allows the identification of sibling Element Complexes, which can occur adjacent to the identified Element Complex in the same depositional system or Element Complex Assemblage.

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